

Online Appendix for: “Belief Distortions, Asset Prices, and Unemployment Fluctuations*”

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OA Online Appendix: Additional Results

OA.1 Variance Decompositions

OA.1.1 Baseline Specification

Table OA.1 reports a variance decomposition of the aggregate vacancy filling rate based on equation (22). Under objective expectations, discount rate fluctuations explain the largest share of variation, accounting for 69.1% at the five-year horizon. Under subjective expectations, cash flow beliefs dominate beyond the one-year horizon, accounting for 96.7% at the five-year horizon.

Table OA.1: Time-Series Decomposition of the Vacancy Filling Rate

Horizon h (Years)	1	2	3	4	5
(a) Objective Expectations: $\log q_t = c_q + \mathbb{E}_t[r_{t,t+h}] - \mathbb{E}_t[e_{t,t+h}] - \mathbb{E}_t[pe_{t,t+h}]$					
Discount Rate	0.187*	0.309**	0.585***	0.653***	0.691***
t -stat	(1.655)	(2.354)	(2.988)	(3.487)	(3.329)
(-) Cash Flow	0.027	0.026	0.051	0.055	0.066
t -stat	(0.091)	(0.181)	(0.365)	(0.459)	(0.472)
(-) Price-Earnings	0.799***	0.720**	0.415	0.331	0.201
t -stat	(2.810)	(2.161)	(0.666)	(0.406)	(0.245)
Residual	-0.013	-0.054	-0.051	-0.039	0.042
t -stat	(-0.030)	(-0.141)	(-0.076)	(-0.046)	(0.049)
N	76	76	76	76	76
(b) Subjective Expectations: $\log q_t = c_q + \mathbb{F}_t[r_{t,t+h}] - \mathbb{F}_t[e_{t,t+h}] - \mathbb{F}_t[pe_{t,t+h}]$					
Discount Rate	-0.007	-0.005	-0.019	-0.014	-0.010
t -stat	(-0.457)	(-0.130)	(-0.400)	(-0.157)	(-0.091)
(-) Cash Flow	0.325**	0.641***	0.717***	0.892***	0.967***
t -stat	(1.969)	(4.501)	(4.661)	(5.572)	(7.097)
(-) Price-Earnings	0.629***	0.366	0.206	0.068	0.028
t -stat	(2.794)	(1.410)	(0.724)	(0.175)	(0.078)
Residual	0.052	-0.002	0.096	0.054	0.015
t -stat	(0.186)	(-0.008)	(0.292)	(0.126)	(0.039)
N	76	76	76	76	76

Notes: This table reports variance decompositions of the aggregate vacancy filling rate under objective expectations (panel (a)) or subjective expectations (panel (b)). Each row reports the share of the variation in vacancy filling rates that can be explained by h -year expected present discounted values of discount rates $r_{t,t+h}$, (negative) cash flows $e_{t,t+h}$, and (negative) price-earnings ratios $pe_{t,t+h}$, as defined in equation (17). Residual term represents the variation in vacancy filling rates that are not captured by the other components. Positive numbers in the Cash Flow and Price-Earnings rows represent the negative of the regression coefficients, ensuring that all variance shares are positive and sum to unity. Subjective expectations $\mathbb{F}_t$ are based on survey forecasts of CFOs and IBES financial analysts. Objective expectations $\mathbb{E}_t$ are based on machine learning forecasts from Long Short-Term Memory (LSTM) neural networks. The sample is quarterly from 2005Q1 to 2023Q4. Newey-West corrected t -statistics with lags = 4 are reported in parentheses: *sig. at 10%. **sig. at 5%. ***sig. at 1%.

OA.1.2 Belief Distortions and Vacancy Filling Rate

To quantify the importance of belief distortions, I regress the belief distortion, the difference between subjective and machine expectations, in discount rate, cash flow, and log price-earnings expectations on the vacancy filling rate (Table OA.2):

$$\mathbb{F}_t[y_{t+h}] - \mathbb{E}_t[y_{t+h}] = \beta_{0,B} + \beta_{1,B} \log q_t + \varepsilon_{t,B}, \quad y = r, e, pe$$

Distortions in survey forecasts are important contributors to vacancy-filling-rate fluctuations, especially at longer horizons. At five years, cash flow distortions lead respondents to over-weight the cash flow component by 90.1% of the variation, partly offset by discount rate distortions that under-weight it by 70.1%. This horizon profile mirrors the MSE ratios in Figure 2: the machine beats the survey by a wider margin at longer horizons, where belief distortions play a bigger role.

Table OA.2: Belief Distortions in Subjective Beliefs and the Vacancy Filling Rate

Horizon h (Years)	1	2	3	4	5
Belief Distortions: $\mathbb{F}_t[y_{t+h}] - \mathbb{E}_t[y_{t+h}] = \beta_{0,B} + \beta_{1,B} \log q_t + \varepsilon_{t,B}, \quad y = r, e, pe$					
Discount Rate	-0.194	-0.313**	-0.604***	-0.667***	-0.701***
t -stat	(-1.574)	(-2.167)	(-2.896)	(-2.918)	(-2.740)
(-) Cash Flow	0.299	0.615***	0.666***	0.837***	0.901***
t -stat	(1.421)	(5.476)	(5.703)	(7.365)	(6.665)
(-) Price-Earnings	-0.170	-0.354**	-0.209	-0.262	-0.174
t -stat	(-0.464)	(-2.373)	(-0.503)	(-0.479)	(-0.292)
Residual	-0.065	-0.052	-0.147	-0.093	0.026
t -stat	(-0.148)	(-0.219)	(-0.306)	(-0.154)	(0.040)
N	76	76	76	76	76

Notes: This table reports estimates $\beta_{1,B}$ from regressing the survey belief distortion $\mathbb{F}_t[y_{t+h}] - \mathbb{E}_t[y_{t+h}]$ on the vacancy filling rate q_t . y_{t+h} denotes the dependent variable of type j to be predicted h years ahead of time t . The components of the decomposition are h -year present discounted values of discount rates $r_{t,t+h}$, (negative) cash flows $e_{t,t+h}$, and (negative) price-earnings ratios $pe_{t,t+h}$. The residual term captures variation in the vacancy filling rate that cannot be explained by the three components. Subjective expectations $\mathbb{F}_t$ are based on survey forecasts from the CFO survey for stock returns, and IBES for earnings growth. Machine expectations are based on machine learning forecasts $\mathbb{E}_t$ from Long Short-Term Memory (LSTM) neural networks $G(\mathcal{X}_t, \beta_{h,t})$, whose parameters $\beta_{h,t}$ are estimated in real time using $\mathcal{X}_t$, a large scale dataset of macroeconomic, financial, and textual data. The belief distortion is defined as the difference between subjective and machine expectations: $\mathbb{F}_t - \mathbb{E}_t$. The sample is quarterly from 2005Q1 to 2023Q4. Newey-West corrected t -statistics with lags = 4 are reported in parentheses: *sig. at 10%. **sig. at 5%. ***sig. at 1%.

OB Model Details

OB.1 Representative Agent Model

In this section, I present a search and matching model based on Diamond (1982), Mortensen (1982), and Pissarides (2009). The model introduces subjective beliefs that may depart from rational expectations, thereby capturing the impact of belief distortions on labor market dynamics. See Petrosky-Nadeau et al. (2018) for a standard search and matching model formulated under rational expectations. I begin with a representative firm setup to develop intuition for the aggregate dynamics, then extend the model in Section OB.2 to include firm heterogeneity to support the cross-sectional analysis. Consider a discrete time economy populated by a representative household and a representative firm that uses labor as a single input to production.

Representative Household The household has a continuum of mass 1 members who are either employed L_t or unemployed U_t at any point in time. The population is normalized to 1, i.e., $L_t + U_t = 1$, meaning that L_t and U_t are also the rates of employment and unemployment, respectively. The household's consumption decision implies a stochastic discount factor M_{t+1} . The household pools the income of all members before making its consumption decision. Assume that the household has perfect consumption insurance and its members have access to complete contingent claims against aggregate risk. Risk sharing implies each member consumes the same amount regardless of idiosyncratic shocks.

Search and Matching At the start of period t , the employment stock L_t reflects the total number of workers carried over from the previous period before any separations or new hires in period t . A fraction δ_t of these workers separate during the period, so the number of continuing employees becomes $(1 - \delta_t)L_t$. The representative firm posts job vacancies V_t and engages in search over the course of the period to attract unemployed workers U_t . Matches are formed at the end of period t according to a matching function $m(U_t, V_t)$, where $q_t \equiv m(U_t, V_t)/V_t$ is the *vacancy filling rate*, and $f_t \equiv m(U_t, V_t)/U_t$ is the *job-finding rate*. These new matches become part of the workforce starting in period $t + 1$, so employment evolves according to the employment accumulation equation:

$$L_{t+1} = (1 - \delta_t)L_t + q_t V_t \quad (\text{OA.1})$$

The vacancy filling rate q_t maps vacancy posting decisions made during period t into employment outcomes observed at the beginning of period $t + 1$. The variance decomposition does not require me to fully specify the matching function m . Posting a vacancy costs the firm $\kappa > 0$ per period, reflecting fixed hiring costs such as training and administrative setup. Jobs are destroyed at a time-varying job separation rate δ_t . Unemployment $U_t = 1 - L_t$ evolves according to:

$$U_{t+1} = \delta_t(1 - U_t) + (1 - q_t \theta_t)U_t \quad (\text{OA.2})$$

where $\theta_t = V_t/U_t$ denotes labor market tightness, defined as the vacancy-to-unemployment ratio.

Representative Firm The firm has access to a production function F which uses labor L_t as an input to produce output $Y_t = F(L_t)$. Dividends to the firm's shareholders E_t are defined as $E_t \equiv \Pi_t - \kappa V_t$, where $\Pi_t \equiv Y_t - W_t L_t$ is the total profit before vacancy posting costs κV_t and W_t is the wage rate. As in Petrosky-Nadeau et al. (2018), I assume that the representative household owns the equity of the firm, and that the firm pays out all of its earnings as dividends. I also assume that firms have the same unconstrained access to financing as investors in the financial market. The firm posts the optimal number of vacancies to maximize the cum-dividend market value of equity S_t :

$$S_t = \max_{\{V_{t+j}, L_{t+j}\}_{j=0}^{\infty}} \mathbb{F}_t \left[\sum_{j=0}^{\infty} M_{t,t+j} E_{t+j} \right] \quad (\text{OA.3})$$

subject to the employment accumulation equation (OA.1). The firm takes the wage rate W_t , household's stochastic discount factor $M_{t,t+j} = \prod_{s=1}^j M_{t+s}$, and vacancy filling rate q_t as given. $\mathbb{F}_t[\cdot]$ denotes expectations conditional on information available at period t , computed based on the firm's possibly distorted beliefs. These beliefs may depart from objective expectations $\mathbb{E}_t[\cdot]$, with the nature and magnitude of the deviation disciplined using survey data.

Hiring Equation The firm's optimal hiring decision equates the expected discounted value of hiring a marginal worker with its marginal cost. Rewrite the firm's problem in equation (OA.3) from infinite-horizon to recursive form:

$$S_t = \max_{V_t, L_{t+1}} \Pi_t - \kappa V_t + \mathbb{F}_t [M_{t+1} S_{t+1}] \quad (\text{OA.4})$$

$$\text{s.t. } L_{t+1} = (1 - \delta_t)L_t + q_t V_t \quad (\text{OA.5})$$

The first-order condition with respect to V_t is:

$$\frac{\partial S_t}{\partial V_t} = -\kappa + \mathbb{F}_t \left[M_{t+1} \frac{\partial S_{t+1}}{\partial L_{t+1}} \frac{\partial L_{t+1}}{\partial V_t} \right] = 0 \quad (\text{OA.6})$$

Substitute $\frac{\partial L_{t+1}}{\partial V_t} = q_t$ and $\frac{\partial L_{t+1}}{\partial L_t} = (1 - \delta_t)$ from the employment accumulation equation (OA.5), and rearrange (OA.6) in terms of the marginal cost of hiring κ/q_t :

$$\frac{\kappa}{q_t} = \mathbb{F}_t \left[M_{t+1} \frac{\partial S_{t+1}}{\partial L_{t+1}} \right] \quad (\text{OA.7})$$

Next, differentiate S_t with respect to L_t :

$$\frac{\partial S_t}{\partial L_t} = \frac{\partial \Pi_t}{\partial L_t} + \mathbb{F}_t \left[M_{t+1} \frac{\partial S_{t+1}}{\partial L_{t+1}} \frac{\partial L_{t+1}}{\partial L_t} \right] \quad (\text{OA.8})$$

Substitute $\frac{\partial L_{t+1}}{\partial L_t} = (1 - \delta_t)$ from the employment accumulation equation (OA.5):

$$\frac{\partial S_t}{\partial L_t} = \frac{\partial \Pi_t}{\partial L_t} + (1 - \delta_t) \mathbb{F}_t \left[M_{t+1} \frac{\partial S_{t+1}}{\partial L_{t+1}} \right] \quad (\text{OA.9})$$

Substitute equation (OA.9) for period $t + 1$ into equation (OA.7):

$$\frac{\kappa}{q_t} = \mathbb{F}_t \left[M_{t+1} \left(\frac{\partial \Pi_{t+1}}{\partial L_{t+1}} + (1 - \delta_{t+1}) \mathbb{F}_{t+1} \left[M_{t+2} \frac{\partial S_{t+2}}{\partial L_{t+2}} \right] \right) \right] \quad (\text{OA.10})$$

Finally, substitute in (OA.7) for period $t + 1$ to arrive at the *hiring equation*:

$$\underbrace{\frac{\kappa}{q_t}}_{\text{Cost of hiring}} = \underbrace{\mathbb{F}_t \left[M_{t+1} \left(\pi_{t+1} + (1 - \delta_{t+1}) \frac{\kappa}{q_{t+1}} \right) \right]}_{\text{Expected discounted value of hiring}} \quad (\text{OA.11})$$

where $\pi_t \equiv \frac{\partial \Pi_t}{\partial L_t}$ is the profit flow from the marginal hired worker. The hiring equation relates the marginal cost of hiring $\frac{\kappa}{q_t}$ with the expected marginal value of hiring to the firm, which equals the future expected marginal benefits of hiring discounted to present value with the stochastic discount factor M_{t+1} . The future marginal benefits of hiring include π_{t+1} , the future marginal product of labor net of the wage rate, plus the future marginal value of hiring, which equals the future marginal cost of hiring $\frac{\kappa}{q_{t+1}}$ net of separation $(1 - \delta_{t+1})$. During recessions, vacancy filling rates q_t are high, which makes the cost of hiring κ/q_t low. The low cost of hiring must be rationalized by either low expected discounted profit flows $\mathbb{F}_t[M_{t+1}\pi_{t+1}]$ or low future value of hiring $(1 - \delta_{t+1})\frac{\kappa}{q_{t+1}}$. The hiring equation is the labor market analogue of the optimality condition for physical capital in the q theory of investment (Hayashi, 1982), where κ/q_t is the upfront cost of investment analogous to Tobin's marginal q and δ_{t+1} is the depreciation rate.

Constant Returns to Scale (CRS) Next, I derive the firm's stock price implied by the optimal hiring decision. Assume a constant returns to scale (CRS) production function so that marginal profits equal average profits:

$$\pi_{t+1} L_{t+1} = \frac{\partial \Pi_{t+1}}{\partial L_{t+1}} L_{t+1} = \Pi_{t+1} \quad (\text{OA.12})$$

Multiply both sides of the hiring equation by the number of employees L_{t+1} :

$$\frac{\kappa}{q_t} L_{t+1} = \mathbb{F}_t \left[M_{t+1} \left(\pi_{t+1} L_{t+1} + (1 - \delta_{t+1}) \frac{\kappa}{q_{t+1}} L_{t+1} \right) \right] \quad (\text{OA.13})$$

Substitute in the employment accumulation equation (OA.5) and rearrange terms:

$$\frac{\kappa}{q_t} L_{t+1} = \mathbb{F}_t \left[M_{t+1} \left(\pi_{t+1} L_{t+1} + \frac{\kappa}{q_{t+1}} (L_{t+2} - q_{t+1} V_{t+1}) \right) \right] \quad (\text{OA.14})$$

$$= \mathbb{F}_t \left[M_{t+1} \left(\pi_{t+1} L_{t+1} - \kappa V_{t+1} + \frac{\kappa}{q_{t+1}} L_{t+2} \right) \right] \quad (\text{OA.15})$$

Use the constant returns to scale assumption to simplify $\pi_{t+1}L_{t+1} - \kappa V_{t+1} = \Pi_{t+1} - \kappa V_{t+1} = E_{t+1}$:

$$\frac{\kappa}{q_t}L_{t+1} = \mathbb{F}_t \left[M_{t+1} \left(E_{t+1} + \frac{\kappa}{q_{t+1}}L_{t+2} \right) \right] \quad (\text{OA.16})$$

Substitute the equation recursively:

$$\frac{\kappa}{q_t}L_{t+1} = \mathbb{F}_t \left[\sum_{j=1}^{\infty} M_{t,t+j} E_{t+j} \right] + \lim_{T \rightarrow \infty} \mathbb{F}_t \left[M_{t,t+T} \frac{\kappa}{q_{t+T}}L_{t+T+1} \right] \quad (\text{OA.17})$$

The first term on the right-hand side is the firm's stock price $P_t \equiv S_t - E_t$, which is the firm's ex-dividend equity value. Take the second term to zero by applying a transversality condition to arrive at an equation that relates the total cost of hiring with the firm's stock price:

$$\frac{\kappa}{q_t}L_{t+1} = P_t \quad (\text{OA.18})$$

where employment L_{t+1} is determined at the end of date t under the timing convention from equation (OA.1). Take logarithms of both sides of the firm's stock price equation (OA.18) and rearrange terms:

$$\log \kappa - \log q_t = \log \frac{P_t}{L_{t+1}} = \log \frac{P_t}{E_t} - \log \frac{E_t}{L_{t+1}} \equiv pe_t - el_t \quad (\text{OA.19})$$

where I define $pe_t \equiv \log \frac{P_t}{E_t}$ and $el_t \equiv \log \frac{E_t}{L_{t+1}}$ for notational convenience.

Log-linear Approximation of Price-Earnings Ratio To express the price-earnings ratio pe_t in terms of forward-looking variables, start by log-linearizing the price-dividend ratio $pd_t = \log(P_t/D_t)$ around its long-term average $\overline{pd}$ (Campbell and Shiller, 1988):

$$pd_t = c_{pd} + \Delta d_{t+1} - r_{t+1} + \rho pd_{t+1} \quad (\text{OA.20})$$

where c_{pd} is a linearization constant, $r_{t+1} \equiv \log(\frac{P_{t+1}+D_{t+1}}{P_t})$ is the log stock return (with dividends), and $\rho \equiv \exp(\overline{pd})/(1 + \exp(\overline{pd})) = 0.98$ is a persistence parameter that arises from the log linearization. Rewrite the equation in terms of log price-earnings instead of log price-dividends by using the identity $pe_t = pd_t + de_t$, where de_t log payout ratio:

$$pe_t = c_{pd} + \Delta e_{t+1} - r_{t+1} + \rho pe_{t+1} + (1 - \rho)de_{t+1} \quad (\text{OA.21})$$

Since $1 - \rho \approx 0$ and the payout ratio de_t is bounded, $(1 - \rho)de_{t+1}$ can be approximated as a constant, i.e., $c_{pe} \approx c_{pd} + (1 - \rho)de_{t+1}$ (De La O et al., 2024):

$$pe_t \approx c_{pe} + \Delta e_{t+1} - r_{t+1} + \rho pe_{t+1} \quad (\text{OA.22})$$

Recursively substitute for the next h periods

$$pe_t = \sum_{j=1}^h \rho^{j-1} (c_{pe} + \Delta e_{t+j} - r_{t+j}) + \rho^h pe_{t+h} \quad (\text{OA.23})$$

Decomposition of Vacancy Filling Rate Substitute the log-linearized price-earnings ratio in equation (OA.23) into the hiring equation in equation (OA.19):

$$\log q_t = \log \kappa - pe_t - el_t = \log \kappa - \left[\sum_{j=1}^h \rho^{j-1} (c_{pe} + \Delta e_{t+j} - r_{t+j}) + \rho^h pe_{t+h} \right] - el_t \quad (\text{OA.24})$$

Rearrange and collect terms to obtain an ex-post decomposition of the vacancy filling rate:

$$\log q_t = \underbrace{c_q + \sum_{j=1}^h \rho^{j-1} r_{t+j}}_{r_{t,t+h}} - \underbrace{\left[el_t + \sum_{j=1}^h \rho^{j-1} \Delta e_{t+j} \right]}_{e_{t,t+h}} - \underbrace{\rho^h pe_{t+h}}_{pe_{t,t+h}} \quad (\text{OA.25})$$

where $c_q \equiv \log \kappa - \frac{c_{pe}(1-\rho^h)}{1-\rho}$ is a constant. The equation decomposes the vacancy filling rate into future discount rates $r_{t,t+h} \equiv \sum_{j=1}^h \rho^{j-1} r_{t+j}$, cash flows $e_{t,t+h} \equiv el_t + \sum_{j=1}^h \rho^{j-1} \Delta e_{t+j}$, and price-earnings $pe_{t,t+h} \equiv \rho^h pe_{t+h}$. The cash flow component consists of one period ahead log earnings-employment el_t , which captures news about current cash flow fluctuations, and $j = 1, \dots, h$ period ahead log earnings growth Δe_{t+j} , which captures news about future cash flows. The earnings-employment ratio can be interpreted as a measure of the marginal product of labor under constant returns to scale (David et al., 2022). $pe_{t,t+h}$ is a terminal value that captures other long-term influences beyond h periods into the future not already captured in discount rates and cash flows. Since equation (OA.25) holds both ex-ante and ex-post, it can be evaluated under either subjective or objective expectations. The *subjective decomposition* replaces ex-post realizations of future outcomes with their subjective expectations:

$$\log q_t = c_q + \underbrace{\sum_{j=1}^h \rho^{j-1} \mathbb{F}_t[r_{t+j}]}_{\mathbb{F}_t[r_{t,t+h}]} - \underbrace{\left[el_t + \sum_{j=1}^h \rho^{j-1} \mathbb{F}_t[\Delta e_{t+j}] \right]}_{\mathbb{F}_t[e_{t,t+h}]} - \underbrace{\rho^h \mathbb{F}_t[pe_{t+h}]}_{\mathbb{F}_t[pe_{t,t+h}]} \quad (\text{OA.26})$$

Alternatively, the *objective decomposition* replaces ex-post realizations of future outcomes with their objective expectations:

$$\log q_t = c_q + \underbrace{\sum_{j=1}^h \rho^{j-1} \mathbb{E}_t[r_{t+j}]}_{\mathbb{E}_t[r_{t,t+h}]} - \underbrace{\left[el_t + \sum_{j=1}^h \rho^{j-1} \mathbb{E}_t[\Delta e_{t+j}] \right]}_{\mathbb{E}_t[e_{t,t+h}]} - \underbrace{\rho^h \mathbb{E}_t[pe_{t+h}]}_{\mathbb{E}_t[pe_{t,t+h}]} \quad (\text{OA.27})$$

Comparing these decompositions can quantify how belief distortions affect the vacancy filling rate.

Estimation The econometrician can estimate the variance decomposition using predictive regressions of each expected outcome on the current vacancy filling rate. For the subjective decomposition, demean each variable in equation (OA.26), multiply both sides by the current log vacancy filling rate $\log q_t$, and take the sample average:

$$\text{Var}[\log q_t] = \text{Cov}[\mathbb{F}_t[r_{t,t+h}], \log q_t] - \text{Cov}[\mathbb{F}_t[e_{t,t+h}], \log q_t] - \text{Cov}[\mathbb{F}_t[pe_{t,t+h}], \log q_t] \quad (\text{OA.28})$$

where $\text{Var}[\cdot]$ and $\text{Cov}[\cdot]$ are sample variances and covariances based on data observed over a historical sample. Finally, divide both sides by $\text{Var}[\log q_t]$ to decompose its variance:

$$1 = \underbrace{\frac{\text{Cov}[\mathbb{F}_t[r_{t,t+h}], \log q_t]}{\text{Var}[\log q_t]}}_{\text{Discount Rate News}} - \underbrace{\frac{\text{Cov}[\mathbb{F}_t[e_{t,t+h}], \log q_t]}{\text{Var}[\log q_t]}}_{\text{Cash Flow News}} - \underbrace{\frac{\text{Cov}[\mathbb{F}_t[pe_{t,t+h}], \log q_t]}{\text{Var}[\log q_t]}}_{\text{Future Price-Earnings News}} \quad (\text{OA.29})$$

The left-hand side represents the full variability in vacancy filling rates, hence is equal to one. Each term on the right reflects the share explained by subjective expectations of discount rates, cash flows, or price-earnings ratios. Under stationarity, the econometrician can estimate these shares using the OLS coefficients from regressing $\mathbb{F}_t[r_{t,t+h}]$, $\mathbb{F}_t[e_{t,t+h}]$, and $\mathbb{F}_t[pe_{t,t+h}]$ on the current log vacancy filling rate $\log q_t$, respectively. Finally, the decomposition under objective expectations can be estimated similarly based on equation (OA.27) by replacing the subjective expectation $\mathbb{F}_t[\cdot]$ with its objective counterpart $\mathbb{E}_t[\cdot]$:

$$1 = \underbrace{\frac{\text{Cov}[\mathbb{E}_t[r_{t,t+h}], \log q_t]}{\text{Var}[\log q_t]}}_{\text{Discount Rate News}} - \underbrace{\frac{\text{Cov}[\mathbb{E}_t[e_{t,t+h}], \log q_t]}{\text{Var}[\log q_t]}}_{\text{Cash Flow News}} - \underbrace{\frac{\text{Cov}[\mathbb{E}_t[pe_{t,t+h}], \log q_t]}{\text{Var}[\log q_t]}}_{\text{Future Price-Earnings News}} \quad (\text{OA.30})$$

Under stationarity, the econometrician can estimate these shares using the OLS coefficients from regressing $\mathbb{E}_t[r_{t,t+h}]$, $\mathbb{E}_t[e_{t,t+h}]$, and $\mathbb{E}_t[pe_{t,t+h}]$ on the current log vacancy filling rate $\log q_t$, respectively.

OB.2 Cross-Sectional Decomposition of Hiring Rate

The cross-sectional analysis employs a firm-level hiring framework that is the direct analogue of the aggregate representative firm search model. Both approaches derive from the same fundamental principle: firms hire until the marginal cost of hiring equals the marginal value of an additional worker. The key difference lies in the level of aggregation and the specific frictions that generate hiring costs. In the aggregate search model, linear vacancy posting costs (κ per vacancy) combined with constant returns to scale imply that marginal value equals average value, leading to the simplified hiring condition $\frac{\kappa}{q_t} = \frac{P_t}{L_{t+1}}$. For cross-sectional analysis, I retain firm-level heterogeneity and introduce convex adjustment costs that generate dispersion in hiring rates while preserving the core economic mechanism linking firm valuations to hiring decisions.

Consider firm i with production function:

$$Y_{i,t} = A_{i,t} L_{i,t}^\alpha \quad (\text{OA.31})$$

where $A_{i,t}$ represents productivity and $L_{i,t}$ is labor input. The firm's earnings, net of hiring costs and wages, are:

$$E_{i,t} = Y_{i,t} - \phi\left(\frac{H_{i,t}}{L_{i,t}}\right) L_{i,t} - W_{i,t} L_{i,t} \quad (\text{OA.32})$$

where $\phi(\cdot)$ captures convex adjustment costs for hiring at rate $H_{i,t}/L_{i,t}$, and $W_{i,t}$ is the equilibrium wage. The adjustment cost function $\phi(\cdot)$ represents the firm-level analogue of the aggregate matching friction. While the search model features linear vacancy costs that aggregate to determine the market-wide vacancy filling rate, individual firms face convex costs when rapidly adjusting their workforce due to capacity constraints in recruitment, training bottlenecks, and organizational frictions. Firm value satisfies the Bellman equation:

$$V(A_{i,t}, L_{i,t}) = \max_{H_{i,t}} \left\{ E_{i,t} + \mathbb{E}_t \left[\frac{M_{t+1}}{M_t} V(A_{i,t+1}, L_{i,t+1}) \right] \right\} \quad (\text{OA.33})$$

subject to the employment accumulation equation:

$$L_{i,t+1} = (1 - \delta_{i,t}) L_{i,t} + H_{i,t} \quad (\text{OA.34})$$

where M_t is the stochastic discount factor and $\delta_{i,t}$ is the job separation rate. The first-order condition with respect to hiring equates marginal cost to marginal benefit:

$$\phi'\left(\frac{H_{i,t}}{L_{i,t}}\right) = \mathbb{E}_t \left[\frac{M_{t+1}}{M_t} \frac{\partial V(A_{i,t+1}, L_{i,t+1})}{\partial L_{i,t+1}} \right] \quad (\text{OA.35})$$

Under constant returns to scale, the envelope theorem yields $\frac{\partial V}{\partial L} = \frac{V}{L}$, allowing me to express the marginal value in terms of observable quantities:

$$\phi'\left(\frac{H_{i,t}}{L_{i,t}}\right) = \mathbb{E}_t \left[\frac{M_{t+1}}{M_t} \frac{V(A_{i,t+1}, L_{i,t+1})}{L_{i,t+1}} \right] = \frac{P_{i,t}}{L_{i,t+1}} \quad (\text{OA.36})$$

where $P_{i,t}$ is the ex-dividend firm value (stock price). This hiring condition is the firm-level equivalent of the aggregate search model's condition $\frac{\kappa}{q_t} = \frac{P_t}{L_{t+1}}$. Both express the fundamental insight that hiring depends on the ratio of firm value to employment, but the cross-sectional version allows for firm-specific variation in both the adjustment cost parameters and the value-to-employment ratios. Assuming quadratic adjustment costs $\phi\left(\frac{H_{i,t}}{L_{i,t}}\right) = \frac{c_l}{2} \left(\frac{H_{i,t}}{L_{i,t}}\right)^2$, the marginal cost becomes $\phi'\left(\frac{H_{i,t}}{L_{i,t}}\right) = c_l \frac{H_{i,t}}{L_{i,t}}$, yielding:

$$c_l \frac{H_{i,t}}{L_{i,t}} = \frac{P_{i,t}}{L_{i,t+1}} \quad (\text{OA.37})$$

Taking logs and decomposing the price-to-employment ratio:

$$\ln(c_l) + \ln\left(\frac{H_{i,t}}{L_{i,t}}\right) = \ln\left(\frac{P_{i,t}}{E_{i,t}}\right) - \ln\left(\frac{E_{i,t+1}}{E_{i,t}}\right) + \ln\left(\frac{E_{i,t+1}}{L_{i,t+1}}\right) \quad (\text{OA.38})$$

Using lowercase letters to denote log variables, this becomes:

$$\ln(c_l) + h_{i,t} = p_{e,i,t} - \Delta e_{i,t+1} + e_{l,i,t+1} \quad (\text{OA.39})$$

Substituting the Campbell-Shiller decomposition of the log price-earnings ratio:

$$p_{e,i,t} \approx c + \sum_{j=1}^h \rho^{j-1} \Delta e_{i,t+j} - \sum_{j=1}^h \rho^{j-1} r_{i,t+j} + \rho^h p_{e,i,t+h} \quad (\text{OA.40})$$

Taking subjective expectations and cross-sectionally demeaning to eliminate common terms yields the final decomposition:

$$\tilde{h}_{i,t} \approx \underbrace{\mathbb{E}_t[\tilde{e}_{l,i,t+1}] + \sum_{j=2}^h \rho^{j-1} \mathbb{E}_t[\Delta \tilde{e}_{i,t+j}]}_{\text{Cash Flow}} - \underbrace{\sum_{j=1}^h \rho^{j-1} \mathbb{E}_t[\tilde{r}_{i,t+j}]}_{\text{Discount Rate}} + \underbrace{\rho^h \mathbb{E}_t[\tilde{p}_{e,i,t+h}]}_{\text{Future Price-Earnings}} \quad (\text{OA.41})$$

where $\tilde{x}_{i,t}$ denotes the cross-sectionally demeaned variable. The variance decomposition follows directly from the hiring rate decomposition:

$$1 \approx \underbrace{\frac{\text{Cov}(\mathbb{E}_t[\tilde{e}l_{i,t,t+h}], \tilde{h}l_{i,t})}{\text{Var}(\tilde{h}l_{i,t})}}_{\text{CF}_h} + \underbrace{\frac{\text{Cov}(-\mathbb{E}_t[\tilde{r}_{i,t,t+h}], \tilde{h}l_{i,t})}{\text{Var}(\tilde{h}l_{i,t})}}_{\text{DR}_h} + \underbrace{\frac{\text{Cov}(\mathbb{E}_t[\tilde{p}e_{i,t,t+h}], \tilde{h}l_{i,t})}{\text{Var}(\tilde{h}l_{i,t})}}_{\text{PE}_h} \quad (\text{OA.42})$$

where $\tilde{e}l_{i,t,t+h} \equiv el_{i,t+1} + \sum_{j=2}^h \rho^{j-1} \Delta \tilde{e}_{i,t+j}$, $\tilde{r}_{i,t,t+h} \equiv \sum_{j=1}^h \rho^{j-1} \tilde{r}_{i,t+j}$, and $\tilde{p}e_{i,t,t+h} \equiv \rho^h \tilde{p}e_{i,t+h}$. These covariance terms are estimated as coefficients from univariate regressions with time fixed effects, allowing me to isolate the cross-sectional variation attributable to each component while controlling for aggregate time-series effects.

The data uses Compustat annual employment data ($L_{i,t}$, variable EMP) from 2000 to 2023. The firm-level hiring rate is constructed from the employment accumulation equation:

$$\frac{H_{i,t}}{L_{i,t}} = \frac{L_{i,t+1}}{L_{i,t}} - (1 - \delta_{i,t}) \quad (\text{OA.43})$$

where the job separation rate $\delta_{i,t}$ uses industry-level data from JOLTS.

OC Data Details

This section describes the time-series and cross-sectional data sources used in the estimation. I use quarterly data on the variables represented in the decomposition from equations (OA.26) and (OA.27): employment L_t , unemployment U_t , vacancy filling rates q_t , stock returns $r_{t,t+h}$, earnings growth $\Delta e_{t,t+h}$, price-earnings ratio pe_{t+h} , and earnings-employment ratio el_{t+h} . For each dependent variable of the decomposition, I also construct their corresponding survey expectations $\mathbb{E}_t$ and machine expectations $\mathbb{E}_t$.

Employment For realized values of employment, I first construct an annual series for the aggregate number of employees (EMP) of the S&P 500 constituents by using accounting information from the CRSP and Compustat Merged Annual Industrial Files. The data spans 1970 to 2023 and was downloaded from WRDS on May 15, 2024. I aggregate the firm-level employment data to construct a total employment series for the S&P 500. I interpolate this series to a monthly frequency by using the fitted values from real-time regressions of log annual Compustat employment series on the log monthly BLS series for total nonfarm payrolls (PAYEMS). The regressions are estimated over recursively expanding samples from an initial monthly sample that begins on 1970:01 and ends on the month of the data release for each month's total nonfarm payrolls. To ensure that the fitted values do not use future information not available on each data release, I align each monthly BLS nonfarm payroll release with the annual Compustat S&P 500 employment series from the previous calendar year. To obtain a measure of employment L_{t+1} at the beginning of period $t+1$, I convert the monthly interpolated values to a quarterly frequency by taking the value of the series as of the last month of each calendar quarter. This timing assumption ensures that the measures are consistent with the timing conventions from Section OB while still remaining known to firms by the end of period t . Data on nonfarm payrolls was downloaded through FRED on May 15, 2024.

Vacancy Filling Rate I construct a monthly series for the number of vacancies V_t following Barnichon (2010), by using JOLTS job openings starting 2000:12 (JTS00000000JOL) and extending the series back in time using the help-wanted index before 2000:12. The vacancies data has been downloaded from the author's website on May 19, 2024. For realized values of unemployment U_t , I use the BLS monthly series for the unemployment level (UNEMPLOY), downloaded through FRED on May 15, 2024. Labor market tightness $\theta_t = V_t/U_t$ is the ratio between vacancies and unemployment. The job separation rate δ_t uses the corresponding series from JOLTS.

I follow Shimer (2012) in constructing the job separation rate δ_t , job-finding rate f_t , and vacancy filling rate q_t . Job separation rate is the share of short-term unemployed out of total employment $\delta_t = U_t^s/L_t$, where U_t^s is the BLS series for the number of unemployed less than 5 weeks (UEMPLT5) that was downloaded through FRED on May 15, 2024. The job-finding rate is:

$$f_t = 1 - \frac{U_t - U_t^s}{U_{t-1}}$$

The expression for the job-finding rate follows from the unemployment accumulation equation:

$$U_t = (1 - f_t)U_{t-1} + U_t^s$$

which states that unemployment U_t consists of either the previously unemployed U_{t-1} who did not find a job $(1 - f_t)$, or the short-term unemployed U_t^s that lost a job during the current period. The vacancy filling rate is defined as the share of filled vacancies $f_t V_t$ out of unemployment U_t :

$$q_t = \frac{f_t}{\theta_t} = \frac{f_t U_t}{V_t}$$

I first construct the vacancy filling rate q_t at the monthly frequency. To remove high-frequency fluctuations that likely reflect measurement errors, I time-aggregate the monthly series to a quarterly frequency by taking a 3-month trailing average that ends on the first month of each calendar quarter. This timing assumption ensures that the survey and machine expectations in the variance decomposition do not use advance information about vacancy filling rates that were not published at the time of each forecast. To ensure that all variables used in the variance decomposition are stationary, I follow Shimer (2012) by detrending the quarterly vacancy filling rate q_t using an HP filter with a smoothing parameter of 10^5 .

Realized Stock Returns Stock market returns use monthly data on CRSP value-weighted returns including dividends (VWRETD) from the Center for Research in Security Prices (CRSP). I compute annualized log stock returns by compounding the monthly returns using $r_{t+h} \equiv \frac{1}{h} \sum_{j=1}^{12h} \log(1 + VWRETD_{t+j/12})$. The data was downloaded from WRDS on May 15, 2024. When evaluating the MSE ratios of the machine relative to that of a benchmark survey, I compute machine forecasts for either annual CRSP returns or S&P 500 price growth depending on which value most closely aligns with the concept that survey respondents are asked to predict. To measure one-year stock market price growth, I use the one-year log cumulative growth rate of the S&P 500 index, $\Delta p_{t+1} \equiv \log(P_{t+1}/P_t)$. The monthly S&P index series spans the period 1957:03 to 2022:12 and was downloaded from WRDS on May 15, 2024 from the Annual Update data of the Index File on the S&P 500.

Survey Expectations of Stock Returns

CFO Survey I use survey forecasts of S&P 500 stock returns from the CFO survey to measure subjective return expectations. The CFO survey is a quarterly survey that asks respondents about their expectations for the S&P 500 return over the next 12 months and 10 years ahead, obtained from https://www.richmondfed.org/-/media/RichmondFedOrg/research/national_economy/cfo_survey/current_historical_cfo_data.xlsx. I use the mean point forecast for the value of the “most likely” future stock return in the estimation. More specifically, the survey asks the respondent “*over the next 12 months, I expect the average annual S&P 500 return will be: Most Likely: I expect the return to be: ___%*”. The survey question for stock return expectations 10 years ahead is “*over the next 10 years, I expect the average annual S&P 500 return will be: Most Likely: I expect the return to be: ___%*”. The CFO survey panel includes firms that range from small operations to Fortune 500 companies across all major industries. Respondents include chief financial officers, owner-operators, vice presidents, and directors of finance, and others with financial decision-making roles. The CFO panel has 1,600 members as of December 2022.

I take a stand on the information set of respondents when each forecast was made, and I assume that respondents could have used all data released before they completed the survey. Because the CFO survey releases quarterly forecasts at the end of each quarter, I conservatively set the response deadline for the machine forecast to be the first day of the last month of each quarter (e.g., March 1st). The data spans the periods 2001Q4 to 2023Q4 and were downloaded on March 20th, 2024. Mean point forecasts before 2020Q3 are available in column `sp.1.exp` of sheet `through.Q1.2020`; mean point forecasts from 2020Q3 and onwards are available in column `sp.12moexp.2` of sheet `CFO.SP500`. The forecast is not available in 2019Q1, 2019Q4, 2020Q1, and 2020Q2. I impute the missing forecast for 2019Q1 by linearly interpolating between the available forecasts from 2018Q4 and 2019Q2. I impute the missing forecasts for 2019Q4, 2020Q1, and 2020Q2 by interpolating with the nearest available forecast between 2019Q3 and 2020Q3. Following Nagel and Xu (2022), I assume that the forecasted S&P 500 return includes dividends and capture expectations about annualized cumulative simple net returns compounded from time t to $t+h$, i.e., $\mathbb{F}_t[R_{t,t+h}]$. To obtain survey expectations of log returns $\mathbb{F}_t[\log(1 + r_{t,t+h})]$ from a survey expectation of net simple returns $\mathbb{F}_t[R_{t,t+h}]$, I use the approximation $\mathbb{F}_t[\log(1 + r_{t,t+h})] \approx \log(1 + \mathbb{F}_t[R_{t,t+h}])$.

To obtain long-horizon survey expectations of annualized cumulative log S&P 500 returns over the next $1 < h < 10$ years, I interpolate the forecasts across annualized 1 year and 10 year cumulative log return expectations:

$$\mathbb{F}_t[r_{t,t+h}] = \frac{10-h}{10-1} \mathbb{F}_t[r_{t,t+1}] + \frac{h-1}{10-1} \mathbb{F}_t[r_{t,t+10}], \quad h = 1, 2, \dots, 10$$

Finally, I use the difference between cumulative long-horizon log return expectations between adjacent years (i.e., $\mathbb{F}_t[r_{t,t+h-1}]$ and $\mathbb{F}_t[r_{t,t+h}]$) to obtain $\mathbb{F}_t[r_{t+h}]$, the survey expectation of forward one-year log stock returns h years ahead:

$$\mathbb{F}_t[r_{t+h}] = h \times \mathbb{F}_t[r_{t,t+h}] - (h-1) \times \mathbb{F}_t[r_{t,t+h-1}], \quad h = 1, 2, \dots, 10$$

IBES and Value Line I proxy expected firm-level stock returns using price growth expectations following De La O et al. (2024). Specifically, I construct expected price growth from IBES 12-month median price targets and Value Line 3–5 year median price targets, interpolating linearly for intermediate horizons.

To construct expected price growth, I combine short- and long-term price targets from two sources. For the short horizon, I use the 12-month median price targets from the Institutional Brokers Estimate System (IBES) database. For longer horizons, I use the median price targets from Value Line, which provide the expected stock price level approximately

3–5 years into the future for each firm. These targets reflect analysts’ consensus expectations for each firm’s stock price. I interpret the Value Line price target as the expected price level five years ahead and interpolate linearly between the IBES 12-month price target and the Value Line five-year price target to construct expected price growth for intermediate horizons between one and five years. For each firm i , expected annualized price growth over horizon h is given by:

$$\mathbb{F}_t[r_{i,t+h}] \approx \frac{1}{h} \log \left(\frac{\mathbb{F}_t[P_{i,t+h}]}{P_{i,t}} \right)$$

where $\mathbb{F}_t[P_{i,t+h}]$ is the forecasted price at horizon h , constructed through linear interpolation of IBES and Value Line targets, and $P_{i,t}$ is the observed stock price at time t . As shown in De La O et al. (2024), using price growth expectations to approximate expected firm-level stock returns is reasonably accurate, as dividends represent a relatively small component of total returns for most firms.

Gallup/UBS Survey The UBS/Gallup is a monthly survey of one-year-ahead stock market return expectations. I use the mean point forecast in my estimation and compare these to machine forecasts of the annual CRSP return. Gallup conducted 1,000 interviews of investors during the first two weeks of every month and results were reported on the last Monday of the month. The first survey was conducted on 1998:05. Until 1992:02, the survey was conducted quarterly on 1998:05, 1998:09, and 1998:11. The data on 1998:06, 1998:07, 1998:08, 1998:10, 1998:12, 1999:01, and 2006:01 are missing because the survey was not conducted on these months. I follow Adam et al. (2021) in starting the sample after 1999:02 due to missing values at the beginning of the sample.

For each month when the survey was conducted, respondents are asked about the return they expect on their own portfolio. The survey question is “*What overall rate of return do you expect to get on your portfolio in the next twelve months?*” Before 2003:05, respondents are also asked about the return they expect from an investment in the stock market during the next 12 months. The survey question is “*Thinking about the stock market more generally, what overall rate of return do you think the stock market will provide investors during the coming twelve months?*” For each month, I calculate the average expectations of returns on their own portfolio and returns on the market index. When calculating the average, survey respondents are weighted by the weight factor provided in the survey. I exclude extreme observations where a respondent reported expected returns higher than 95% or lower than -95%.

In order to construct a consistent measure of stock market return expectations over the entire sample period, I impute missing market return expectations using the fitted values from two regressions. First, I impute missing values during 1999:02–2005:12 and 2006:02–2007:10 with the fitted value from regressing expected market returns on own portfolio expectations contemporaneously, where the regression is estimated using the part of the sample where both are available. Second, I impute the one missing observation in both market and own portfolio return expectations for 2006:01 with the fitted value from regressing the market return expectations on the lagged own portfolio return expectations, where the coefficients are estimated using part of the sample where both are available, and the fitted value combines the estimated coefficients with lagged own portfolio expectations data from 2005:12. Following Nagel and Xu (2022), I assume that the forecasted stock market return includes dividends and capture expectations about annual simple net stock returns $\mathbb{F}_t[R_{t+1}]$. To obtain survey expectations of annual log returns $\mathbb{F}_t[\log(1 + r_{t+1})]$ from a survey expectation of annual net simple returns $\mathbb{F}_t[R_{t+1}]$, I use the approximation $\mathbb{F}_t[\log(1 + r_{t+1})] \approx \log(1 + \mathbb{F}_t[R_{t+1}])$. After applying all the procedures, the Gallup market return expectations series spans the periods 1999:02 to 2007:10. The data were downloaded on August 1st, 2024 from Roper iPoll: <http://ropercenter.cornell.edu/ubs-index-investor-optimism/>.

I take a stand on the information set of respondents when each forecast was made, and I assume that respondents could have used all data released before they completed the survey. Since interviews are in the first two weeks of a month (e.g., February), I conservatively set the response deadline for the machine forecast to be the first day of the survey month (e.g., February 1st), implying that I allow the machine to use information only up through the end of the previous month (e.g., through January 31st). This ensures that the machine only sees information that would have been available to all UBS/Gallup respondents for that survey month (February). This approach is conservative in the sense that it handicaps the machine, since all survey respondents who are being interviewed during the next month would have access to more timely information than the machine. Since the survey asks about the “one-year-ahead” I interpret the question to be asking about the forecast period spanning from the current survey month to the same month one year ahead.

Michigan Survey of Consumers (SOC) The SOC contains approximately 50 core questions, and a minimum of 500 interviews are conducted by telephone over the course of the entire month, each month. Table 20 of the SOC reports the probability of an increase in stock market in next year. The survey question was “*The next question is about investing in the stock market. Please think about the type of mutual fund known as a diversified stock fund. This type of mutual fund holds stock in many different companies engaged in a wide variety of business activities. Suppose that tomorrow someone were to invest one thousand dollars in such a mutual fund. Please think about how much money this investment would be worth one year from now. What do you think the percent chance that this one thousand dollar investment will increase in value in the year ahead, so that it is worth more than one thousand dollars one year from now?*” When using this survey forecast to compare to machine forecasts, I impute a point forecast for stock market returns using the method described in Section OC below. I compare the imputed point forecast to machine forecasts of CRSP returns.

For the SOC, interviews are conducted monthly typically over the course of an entire month. (In rare cases, interviews may commence at the end of the previous month, as in February 2018 when interviews began on January 31st 2018.) I take a stand on the information set of respondents when each forecast was made, and I assume that respondents could have used all data released before they completed the survey. Since interviews are almost always conducted over the course of an entire month (e.g., February), I conservatively set the response deadline for the machine forecast to be the first day of the survey month (e.g., February 1st), implying that I allow the machine to use information only up through the end of the previous month (e.g., through January 31st). This ensures that the machine only sees information that would have been available to all respondents for that survey month (February). This approach is conservative in the sense that it handicaps the machine, since all survey respondents who are being interviewed during the next month would have access to more timely information than the machine. Since the survey asks about the “year ahead” I interpret the question to be asking about the forecast period spanning the period running from the current survey month to the same month one year ahead. The data spans 2002:06 to 2023:12. The SOC responses were obtained from <https://data.sca.isr.umich.edu/data-archive/mine.php> and downloaded on May 15, 2024.

Livingston Survey Stock Price Forecast I obtain the Livingston Survey S&P 500 index forecast (SPIF) from the Federal Reserve Bank of Philadelphia, and use the mean values in my structural and forecasting models. I compare the one-year growth in these forecasts to machine forecasts of S&P 500 price growth. My sample spans 1947:06 to 2023:06. The forecast series were downloaded on January 24, 2024.

The survey provides semi-annual forecasts on the level of the S&P 500 index. Participants are asked to provide forecasts for the level of the S&P 500 index for the end of the current survey month, 6 months ahead, and 12 months ahead. I use the mean of the respondents’ forecasts each period, where the sample is based on about 50 observations. Most of the survey participants are professional forecasters with “formal and advanced training in economic theory and forecasting and use econometric models to generate their forecasts.” Participants receive questionnaires for the survey in May and November, after the Consumer Price Index (CPI) data release for the previous month. All forecasts are typically submitted by the end of the respective month of May and November. The results of the survey are released near the end of the following month, on June and December of each calendar year. The exact release dates are available on the Philadelphia Fed website, at the header of each news release. I take a stand on the information set of the respondents when each forecast was made by assuming that respondents could have used all data released before they completed the survey. Since all forecasts are typically submitted by the end of May and November of each calendar year, I set the response deadline for the machine forecast to be the first day of the last month of June and December, implying that I allow the machine to use information only up through the end of the May and November.

I follow Nagel and Xu (2021) in constructing one-year stock price growth expectations from the level forecasts. Starting from June 1992, I use the ratio between the 12-month level forecast ($SPIF_{12M_t}$) and 0-month level nowcasts ($SPIF_{ZM_t}$) of the S&P 500 index. Before June 1992, the 0-month nowcast is not available. Therefore I use the annualized ratio between the 12-month ($spi12_t$) and 6-month ($spi6_t$) level forecast of the S&P 500 index

$$\mathbb{F}_t^{(Liv)} \left[\frac{P_{t+1}}{P_t} \right] \approx \begin{cases} \frac{\mathbb{F}_t^{(Liv)}[P_{t+1}]}{\mathbb{F}_t^{(Liv)}[P_t]} = \frac{SPIF_{12M_t}}{SPIF_{ZM_t}} & \text{if } t \geq 1992M6 \\ \left(\frac{\mathbb{F}_t^{(Liv)}[P_{t+1}]}{\mathbb{F}_t^{(Liv)}[P_{t+6}]} \right)^2 = \left(\frac{spi12_t}{spi6_t} \right)^2 & \text{if } t < 1992M6 \end{cases}$$

where P_t is the S&P 500 index and t indexes the survey’s response deadline. To obtain a survey expectation of the log change in price growth I use the approximation $\mathbb{F}_t(\Delta p_{t+1}) \approx \log(\mathbb{F}_t[P_{t+1}]) - \log(P_t)$.

Conference Board (CB) Survey Respondents provide the categorical belief of whether they expect stock prices to “increase,” “decrease,” or stay the “same” over the next year. Since the survey asks respondents about stock prices in the “year ahead,” I interpret the question to be asking about the forecast period from the end of the current survey month to the end of the same month one year ahead. When I use this qualitative survey forecast to compare to machine forecasts, I impute a point forecast for stock market returns using the method described in Section OC below. I compare the imputed point forecast to machine forecasts of CRSP returns.

The survey is conducted monthly and I use the survey responses over 1987:04 to 2022:08. The data was downloaded on September 26, 2022. The survey uses an address-based mail sample design. Questionnaires are mailed to households on or about the first of each month. Survey responses flow in throughout the collection period, with the sample close-out for preliminary estimates occurring around the 18th of the month. Any responses received after then are used to produce final estimates for the month, which are published with the following month’s data. Conversations with those knowledgeable about the survey suggested that most panelists respond early. Any responses received after around the 20th of the month—regardless of when they are filled out—are included in the final (but not preliminary) numbers.

I take a stand on the information set of the respondents when each forecast was made by assuming that respondents could have used all data released before they completed the survey. Since questionnaires reach households on or about the first of each month (e.g., February 1st) and most respondents respond early, I conservatively set the response deadline for the machine forecast to be the first day of the survey month (e.g., February 1st), implying that I allow the machine to use information only up through the end of the previous month (e.g., January 31st).

Converting Qualitative Forecasts to Point Forecasts (SOC and CB) I use the SOC probability to impute a quantitative point forecast of stock returns using a linear regression of CFO point forecasts for returns onto the SOC probability of a price increase. The SOC asks respondents about the percent chance that an investment will “increase in value in the year ahead.” I interpret this as asking about the ex dividend value, i.e., about price growth. The CFO survey is conducted quarterly, where the survey quarters span 2001Q4 to 2021Q1. The SOC survey is conducted monthly, where survey months span 2002:06 to 2021:12. Since the CFO is a quarterly survey, the regression is estimated in real-time over a quarterly overlapping sample. Since the CFO survey is conducted during the last month of the quarter while the SOC is conducted monthly, I align the survey months between CFO and SOC by regressing the quarterly CFO survey point forecast with the qualitative SOC survey response during the last month of the quarter.

Since the SOC survey question is interpreted as asking about S&P 500 price growth while the CFO survey question asks about stock returns including dividends, I follow Nagel and Xu (2021) in subtracting the current dividend yield of the CRSP value weighted index from the CFO variable before running the regression. After estimating the regression, I then add back the dividend yield to the fitted value to obtain an imputed SOC point forecast of stock returns including dividends. Specifically, at time t , I assume that the CFO forecast of stock returns, $\mathbb{F}_t^{\text{CFO}}[r_{t,t+1}]$, minus the current dividend yield, D_t/P_t , is related to the contemporaneous SOC probability of an increase in the stock market next year, $P_{t,t+1}^{\text{SOC}}$, by:

$$\mathbb{F}_t^{\text{CFO}}[r_{t,t+1}] - D_t/P_t = \beta_0 + \beta_1 P_{t,t+1}^{\text{SOC}} + \epsilon_t.$$

The final imputed SOC point forecast is constructed as $\mathbb{F}_t^{\text{SOC}}[r_{t,t+1}] = \hat{\beta}_0 + \hat{\beta}_1 P_{t,t+1}^{\text{SOC}} + D_t/P_t$. I first estimate the coefficients of the above regression over an initial overlapping sample of 2002Q2 to 2004Q4, where the quarterly observations from the CFO survey is regressed on the SOC survey responses from the last month of each calendar quarter. Using the estimated coefficients and the SOC probability from 2005:03 gives the point forecast of the one-year stock return from 2005Q1 to 2006Q1. I then re-estimate this equation, recursively, adding one quarterly observation to the end of the sample at a time, and storing the fitted values. This results in a time series of SOC point forecasts $\mathbb{F}_t^{\text{SOC}}[r_{t,t+1}]$ spanning 2005Q1 to 2021Q1.

The same procedure is done for the Conference Board Survey, except I replace $P_{t,t+1}^{\text{SOC}}$ by $P_{t,t+1}^{\text{CB}}$, a ratio of the proportion of those who respond with “increase” to the sum of “decrease” and “same.” The CB survey asks respondents to provide the categorical belief of whether they expect stock prices to “increase,” “decrease,” or stay the “same” over the next year. I interpret this as asking about price growth. Since the CB survey question is interpreted as asking about S&P 500 price growth while the CFO survey question asks about stock returns including dividends, I follow Nagel and Xu (2021) in subtracting the current dividend yield of the CRSP value weighted index from the CFO variable before running the regression. After estimating the regression, I then add back the dividend yield to the fitted value to obtain an imputed CB point forecast of stock returns including dividends.

The CFO survey is conducted quarterly, where the survey quarters span 2001Q4 to 2021Q1. The CB survey is conducted monthly, where survey months span 1987:04 to 2022:08. The regression is first estimated over an initial overlapping sample of 2001Q4 to 2004Q4, where the quarterly observations from the CFO survey is regressed on the CB survey responses from the last month of each calendar quarter. Using the estimated coefficients and the CB survey response $P_{t,t+1}^{\text{CB}}$ from 2005:03 gives the point forecast of the stock return from 2005Q1 to 2006Q1. I then re-estimate this equation, recursively, adding one observation to the end of the sample at a time, and storing the fitted values. This results in a time series of CB point forecasts $\mathbb{F}_t^{\text{CB}}[r_{t,t+1}]$ over 2005Q1 to 2021Q1.

Nagel and Xu Individual Investor Expectations Nagel and Xu (2021)’s individual investor expectations series for returns covers 1972-1977 (Annual) and 1987Q2–2023Q4 (Quarterly) and combine data from the following surveys:

1. UBS/Gallup: 1998:06-2007:10; Survey captures respondents’ expected stock market returns, in percent, over a 1-year horizon.
2. Michigan Survey of Consumers (SOC): 2002:04-2023:12; Respondents provide the probability of a rise in the stock market over a 1-year horizon.
3. Conference Board (CB): 1987:04-2022:08; Respondents provide the categorical opinion whether they expect stock prices to rise, or stay about where they are, or decline over the next year.
4. Vanguard Research Initiative (VRI): 2014:08; Survey captures respondents’ expected stock market returns, in percent, over a 1-year horizon.
5. Roper: 1974-1977, annual, observed June of each calendar year; Respondents provide the categorical opinion whether they expect stock prices to rise, or stay about where they are, or decline over the next year.
6. Lease, Lewellen, and Schlarbaum (1974, 1977): 1972-1973, annual, observed July of each calendar year; Survey captures respondents’ expected stock market returns, in percent, over a 1-year horizon.

Among these sources, UBS/Gallup and VRI provide direct, point forecasts of expected stock returns, while SOC, CB, and Roper offer qualitative or probabilistic information that requires conversion to consistent return expectations. Nagel and Xu (2021) construct their final series using the following procedure:

1. Start with UBS/Gallup for 1998:06-2007:10 and VRI for 2014:08 since they capture the respondents' expected stock returns relatively closely (other surveys only provide qualitative measures).
2. Regress SOC on UBS/Gallup and VRI using periods of overlapping coverage (2002:04-2007:10). Use the fitted values from this regression to impute missing data for 2007:11-2023:12 (excluding 2014:08).
3. Regress CB on UBS/Gallup and VRI using periods of overlapping coverage (1998:06-2007:10). Use the fitted values from this regression to impute missing data for 1987:04-1998:05 (using CB) and 1974-1977 (using Roper).
4. Use the coefficients from regressing CB on UBS/Gallup and VRI (from step 3) to compute fitted values that convert the probabilistic forecast from Roper into point forecasts of stock returns.
5. Convert expected returns to expected excess returns by subtracting the average 1-year Treasury yield measured at the beginning of the survey month.
6. Aggregate monthly series to a quarterly frequency by taking the average expectation within calendar quarters.

Risk-Free Rates

Realized Risk-Free Rates As a measure of realized risk-free rates r_t^f , I obtain daily series for the annualized three-month Treasury bill rate (DTB3), downloaded from FRED on May 15, 2024. To match the definition used as the target variable in the Survey of Professional Forecasters (SPF), I time-aggregate the daily realized risk-free rate series to a quarterly frequency by taking the quarterly average, as discussed below.

Survey Expectations of Risk-Free Rates I obtain subjective expectations about risk-free rates from median forecasts for the annualized three-month Treasury bill rate from the Survey of Professional Forecasters (SPF). The SPF provides forecasts at the one and ten year horizons. For one year ahead forecasts (TBILL), respondents are asked to provide quarterly forecasts of the quarterly average three-month Treasury bill rate, in percentage points, where the forecasts are for the quarterly average of the underlying daily levels. I interpret the survey to be asking about annual net simple rates $\mathbb{F}_t[R_{t,t+1}^f]$, and approximate the expected log risk-free rate as $\mathbb{F}_t[r_{t,t+1}^f] \approx \log(1 + \mathbb{F}_t[R_{t,t+1}^f])$. For ten year ahead forecasts (BILL10), respondents are asked to provide forecasts for the annual-average rate of return to three-month Treasury bills over the next 10 years, in percentage points. The ten year ahead forecasts are available only for surveys conducted in the first quarter of each calendar year. I interpret the survey to be asking about annualized cumulative net simple rates compounded from the survey quarter to the same quarter that is ten years after the survey year $\mathbb{F}_t[R_{t,t+10}^f]$, and approximate the expected log risk-free rate as $\mathbb{F}_t[r_{t,t+10}^f] \approx \log(1 + \mathbb{F}_t[R_{t,t+10}^f])$. To obtain long-horizon survey expectations of annualized log three-month Treasury bill rates over the next $1 < h < 10$ years, I interpolate the forecasts across annualized 1 year and 10 year return expectations:

$$\mathbb{F}_t[r_{t,t+h}^f] = \frac{10-h}{10-1} \mathbb{F}_t[r_{t,t+1}^f] + \frac{h-1}{10-1} \mathbb{F}_t[r_{t,t+10}^f], \quad h = 1, 2, \dots, 10$$

Finally, I use the difference between the cumulative annualized long-horizon log three-month Treasury bill rate expectations between adjacent years (i.e., $\mathbb{F}_t[r_{t,t+h-1}^f]$ and $\mathbb{F}_t[r_{t,t+h}^f]$) to obtain $\mathbb{F}_t[r_{t,t+h}^f]$, the time t survey expectation of annualized forward log three-month Treasury bill rate h years ahead:

$$\mathbb{F}_t[r_{t+h}^f] = h \times \mathbb{F}_t[r_{t,t+h}^f] - (h-1) \times \mathbb{F}_t[r_{t,t+h-1}^f], \quad h = 1, 2, \dots, 10$$

The surveys are sent out at the end of the first month of each quarter, and collected in the second or third week of the middle month of each quarter. When constructing machine learning forecasts for the risk-free rate, I assume that forecasters could have used all data released before the survey deadlines for the SPF, which are posted online at the Federal Reserve Bank of Philadelphia website. Since surveys are typically sent out at the end of the first month of each quarter, I make the conservative assumption that respondents only had data released by the first day of the second month of each quarter.

Realized Earnings I use IBES street earnings per share (EPS) data that start in 1983:Q4 as the forecast target for IBES analysts. Following the recommendation of Hillenbrand and McCarthy (2024), I use Street earnings as the forecast target for IBES analysts. Street earnings differ from GAAP earnings by excluding discontinued operations, extraordinary charges, and other non-operating items. According to the IBES user guide, analysts submit forecasts after backing out these transitory components, and IBES constructs the realized series to align with those forecasts. While analysts have some discretion over which items to exclude, Hillenbrand and McCarthy (2024) demonstrate that the target of these forecasts corresponds closely to earnings before special items in Compustat, suggesting that street earnings accurately reflect the measure analysts are targeting. To convert EPS to total earnings, I multiply the resulting quarterly EPS series by the quarterly S&P 500 divisor, available at: https://ycharts.com/indicators/sp_500_divisor. The final quarterly total earnings series spans the period 1983:Q4 to 2023:Q4. To extend the sample back to 1965Q1, I use quarterly Compustat data on earnings before special items. As noted in Hillenbrand and McCarthy (2024), this measure closely tracks IBES street earnings, indicating it accurately reflects analysts' forecast targets. IBES street earnings data and Compustat data have been downloaded from WRDS on July 19, 2025. The divisor data were downloaded on July 21, 2025.

Survey Expectations of Earnings I obtain monthly survey data for the median analyst earnings per share forecast and actual earnings per share from the Institutional Brokers Estimate System (IBES) via Wharton Research Data Services (WRDS). The data spans the period 1976:01 to 2023:12.

Short-Term Growth (STG) Expectations I build measures of aggregate S&P 500 earnings expectations growth using the constituents of the S&P 500 at each point in time following De La O and Myers (2021). I first construct expected earnings expectations for aggregate earnings h -months-ahead as:

$$\mathbb{F}_t[E_{t+h}] = \Omega_t \sum_{i \in x_t} \mathbb{F}_t[EPS_{i,t+h}] \cdot S_{i,t} / Divisor_t$$

where $\mathbb{F}$ is the median analyst survey forecast, E is aggregate S&P 500 earnings, EPS_i is earning per share of firm i among all S&P 500 firms x_t for which I have forecasts in IBES for $t+h$, S_i is shares outstanding of firm i , and $Divisor_t$ is calculated as the S&P 500 market capitalization divided by the S&P 500 index. I obtain the number of outstanding shares for all companies in the S&P500 from Compustat. IBES estimates are available for most but not all S&P 500 companies. Following De La O and Myers (2021), I multiply this aggregate by Ω_t , a ratio of total S&P 500 market value to the market value of the forecasted companies at t to account for the fact that IBES does not provide earnings forecasts for all firms in the S&P 500 in every period.

IBES database contains earning forecasts up to five annual fiscal periods (FY1 to FY5) and as a result, I interpolate across the different horizons to obtain the expectation over the next 12 months. This procedure has been used in the literature, including De La O and Myers (2021). Specifically, if the fiscal year of firm XYZ ends nine months after the survey date, I have a 9-month earning forecast $\mathbb{F}_t[E_{t+9}]$ from FY1 and a 21-month forecast $\mathbb{F}_t[E_{t+21}]$ from FY2. I then obtain the 12-month ahead forecast by interpolating these two forecasts as follows,

$$\mathbb{F}_t[E_{t+12}] = \frac{9}{12}\mathbb{F}_t[E_{t+9}] + \frac{3}{12}\mathbb{F}_t[E_{t+21}].$$

To convert the monthly forecast to quarterly frequency, I use the forecast made in the middle month of each quarter, and construct one-year earnings expectations from 1976Q1 to 2023Q4, and the earnings-expectation growth is calculated as an approximation following De La O and Myers (2021):

$$\mathbb{F}_t(\Delta e_{t+12}) \approx \ln(\mathbb{F}_t[E_{t+12}]) - e_t$$

where e_t is log earnings for S&P 500 at time t calculated as $e_t = \log(EPS_t \cdot Divisor_t)$, where EPS_t is the earnings per share for the S&P 500 obtained from Shiller's data depository and S&P Global, as described above.

Long-Term Growth (LTG) Expectations I construct long term expected earnings growth (LTG) for the S&P 500 following Bordalo et al. (2019). Specifically, I obtain the median firm-level LTG forecast from IBES, and aggregate the value-weighted firm-level forecasts,

$$LTG_t = \sum_{i=1}^S LTG_{i,t} \frac{P_{i,t} Q_{i,t}}{\sum_{i=1}^S P_{i,t} Q_{i,t}}$$

where S is the number of firms in the S&P 500 index, and where $P_{i,t}$ and $Q_{i,t}$ are the stock price and the number of shares outstanding of firm i at time t , respectively. $LTG_{i,t}$ is the median forecast of firm i 's long term expected earnings growth. The data spans the periods from 1981:12 to 2023:12. All data were downloaded on July 19, 2025.

Finally, I use the difference between survey expectations of log earnings between adjacent years (i.e., $\mathbb{F}_t[e_{t+h-1}]$ and $\mathbb{F}_t[e_{t+h}]$) to obtain $\mathbb{F}_t[\Delta e_{t+h}] = \mathbb{F}_t[e_{t+h}] - \mathbb{F}_t[e_{t+h-1}]$, the time t survey expectation of forward one-year log earnings growth

$h = 1, 2$ years ahead. For the $h = 3, 4, 5$ year horizon, I interpret the IBES’s Long-Term Growth (LTG) forecast as the forward annual log earnings growth:

$$\mathbb{F}_t[\Delta e_{t+h}] = \begin{cases} \mathbb{F}_t[e_{t+h}] - \mathbb{F}_t[e_{t+h-1}] & \text{if } h = 1, 2 \text{ years} \\ LTG_t & \text{if } h = 3, 4, 5 \text{ years} \end{cases}$$

To estimate any biases in IBES analyst forecasts, the dynamic machine algorithm takes as an input a likely date corresponding to information analysts could have known at the time of their forecast. IBES does not provide an explicit deadline for their forecasts to be returned. Therefore I instead use the “statistical period” day (the day when the set of summary statistics was calculated) as a proxy for the deadline. I set the machine deadline to be the day before this date. The statistical period date is typically between day 14 and day 20 of a given month, implying that the machine deadline varies from month to month. As the machine learning algorithm uses mixed-frequency techniques adapted to quarterly sampling intervals, while the IBES forecasts are monthly, I compare machine and IBES analyst forecasts as of the middle month of each quarter, considering 12-month ahead forecast from the beginning of the month following the survey month.

Price-Earnings Ratio I construct a quarterly series for the price-earnings ratio $PE_t \equiv P_t/E_t$ using the end-of-quarter S&P 500 stock price index P_t and the S&P 500 quarterly total earnings E_t . I infer subjective expectations of the log price-earnings ratio $\mathbb{F}_t[pe_{t+h}]$ by combining the current log price-earnings ratio pe_t with h year ahead subjective expectations of annual log stock returns $\mathbb{F}_t[r_{t+h}]$ and annual log earnings growth $\mathbb{F}_t[\Delta e_{t+h}]$, following the approach used in De La O and Myers (2021). Rearrange the Campbell and Shiller (1988) present value identity for the price-earnings ratio in equation (OA.23) to express the future log price-earnings ratio as a function of current log price-earnings, log earnings growth, and log stock returns:

$$pe_{t+h} = \frac{1}{\rho^h} pe_t - \frac{1}{\rho^h} \sum_{j=1}^h \rho^{j-1} (c_{pe} + \Delta e_{t+j} - r_{t+j})$$

where the equation holds both ex-ante and ex-post. Apply subjective expectations $\mathbb{F}_t$ on both sides of the equation:

$$\mathbb{F}_t[pe_{t+h}] = \frac{1}{\rho^h} pe_t - \frac{1}{\rho^h} \sum_{j=1}^h \rho^{j-1} (c_{pe} + \underbrace{\mathbb{F}_t[\Delta e_{t+j}]}_{\text{Survey (IBES)}} - \underbrace{\mathbb{F}_t[r_{t+j}]}_{\text{Survey (CFO)}}) \quad (\text{OA.44})$$

where subjective expectations about j years ahead forward annual log stock returns $\mathbb{F}_t[r_{t+j}]$ and forward annual log earnings growth $\mathbb{F}_t[\Delta e_{t+j}]$ use survey forecasts from the CFO survey and IBES, respectively. I construct firm-level price-earnings expectations by applying the same log-linear approximation to firm-level expectations of stock returns (from IBES and Value Line) and earnings growth (from IBES).

Earnings-Employment Ratio The current earnings-employment ratio is defined as $EL_t \equiv E_t/L_{t+1}$, where E_t denotes quarterly total earnings for the S&P 500 and L_{t+1} is the employment stock at the beginning of period $t + 1$. I measure L_{t+1} using end-of-period employment levels within each quarter. This timing assumption ensures that the measures are consistent with the timing conventions from Section OB while still remaining known to firms by the end of period t .

Machine Learning Forecasts For each survey forecast, I also construct their corresponding machine learning forecast by estimating a Long Short-Term Memory (LSTM) neural network:

$$\mathbb{E}_t[y_{t+h}] = G(\mathcal{X}_t, \beta_{h,t})$$

where y_{t+h} denotes the variable y to be predicted h years ahead of time t , and $\mathcal{X}_t$ is a large input dataset of right-hand-side variables including the intercept. $G(\mathcal{X}_t, \beta_{h,t})$ denotes predicted values from a LSTM neural network that can be represented by a (potentially) high-dimensional set of finite-valued parameters $\beta_{h,t}$. The machine learning model is estimated using an algorithm that takes into account the data-rich environment in which firms operate (Bianchi et al., 2022 and Bianchi et al., 2024). When constructing machine learning forecasts of each variable, I allow the machine to use only information that would have been available to all survey respondents at the time of each forecast. See Section OD for details about the machine learning algorithm and predictor variables. Machine expectations about the price-earnings ratio $\mathbb{E}_t[pe_{t+h}]$ is constructed similarly to the survey counterpart, by replacing the survey forecasts of stock returns and earnings growth on the right-hand side of equation (OA.44) with the corresponding machine learning forecasts.

For the cross-sectional decomposition, I construct analogous machine learning forecasts of returns, earnings growth, and price-earnings ratios at the firm level using the same LSTM framework, applied to portfolio-specific predictors and outcomes. To keep the machine learning algorithm tractable, I re-estimate the model parameters and update the hyper-parameter cross-validation every four quarters.

OD Machine Learning

OD.1 Machine Algorithm Details

The basic dynamic algorithm follows the six step approach of Bianchi et al. (2022) of 1. Sample partitioning, 2. In-sample estimation, 3. Training and cross-validation, 4. Grid reoptimization, 5. Out-of-sample prediction, and 6. Roll forward and repeat. I refer the interested reader to that paper for details and discuss details of the implementation here only insofar as they differ. At time t , a prior sample of size $\hat{T}$ is partitioned into two subsample windows: a *training sample* consisting of the first T_E observations, and a hold-out *validation sample* of T_V subsequent observations so that $\hat{T} = T_E + T_V$. The training sample is used to estimate the model subject to a specific set of tuning parameter values, and the validation sample is used for tuning the hyperparameters. The model to be estimated over the training sample is

$$y_{t+h} = G^e(\mathcal{X}_t, \beta_{h,t}) + \epsilon_{t+h}.$$

where y_{t+h} is a time series indexed by j whose value in period $h \geq 1$ the machine is asked to predict at time t , $\mathcal{X}_t$ is a large input dataset of right-hand-side variables including the intercept, and $G^e(\cdot)$ is a machine learning estimator that can be represented by a (potentially) high-dimensional set of finite-valued parameters $\beta_{h,t}^e$. I consider two estimators for $G^e(\cdot)$: Elastic Net $G^{\text{EN}}(\mathcal{X}_t, \beta_{j,h}^{\text{EN}})$, and Long Short-Term Memory (LSTM) network $G^{\text{LSTM}}(\mathcal{X}_t, \beta_{j,h}^{\text{LSTM}})$. The $e \in \{\text{EN}, \text{LSTM}\}$ superscripts on β indicate that the parameters depend on the estimator being used (See the next section for a description of EN and LSTM). $\mathcal{X}_t$ always denotes the most recent data that would have been in real time prior to the date on which the forecast was submitted. To ensure that the effect of each variable in the input vector is regularized fairly during the estimation, I standardize the elements of $\mathcal{X}_t$ such that sample means are zero and sample standard deviations are unity. It should be noted that the most recent observation on the left-hand-side is generally available in real time only with a one-period lag, thus the forecasting estimations can only be run with data over a sample that stops one period later than today in real time. The parameters $\beta_{h,t}^e$ are estimated by minimizing the mean-square loss function over the training sample with L_1 and L_2 penalties

$$L(\beta_{h,t}^e, \mathbf{X}_{T_E}, \lambda_t^e) \equiv \underbrace{\frac{1}{T_E} \sum_{\tau=1}^{T_E} (y_{\tau+h} - G^e(\mathcal{X}_\tau, \beta_{h,\tau}^e))^2}_{\text{Mean Square Error}} + \underbrace{\lambda_{1,t}^e \sum_{k=1}^K |\beta_{j,h,t,k}^e|}_{L_1 \text{ Penalty}} + \underbrace{\lambda_{2,t}^e \sum_{k=1}^K (\beta_{j,h,t,k}^e)^2}_{L_2 \text{ Penalty}}$$

where $\mathbf{X}_{T_E} = (y_{t-T_E}, \dots, y_t, \mathcal{X}'_{t-T_E}, \dots, \mathcal{X}'_t)'$ is the vector containing all observations in the training sample of size T_E . The estimated $\beta_{h,t}^e$ is a function of the data $\mathbf{X}_{T_E}$ and a non-negative regularization parameter vector $\lambda_t^e = (\lambda_{1,t}^e, \lambda_{2,t}^e, \lambda_{0,t}^{\text{LSTM}})'$ where $\lambda_{0,t}^{\text{LSTM}}$ is a set of hyperparameters only relevant when using the LSTM estimator for $G^e(\cdot)$ (see below). For the EN case there are only two hyperparameters, which determine the optimal shrinkage and sparsity of the time t machine specification. The regularization parameters λ_t^e are estimated by minimizing the mean-square loss over pseudo-out-of-sample forecast errors generated from rolling regressions through the validation sample:

$$\hat{\lambda}_t^e, \hat{T}_E, \hat{T}_V = \underset{\lambda_{t,T_E,T_V}^e}{\operatorname{argmin}} \left\{ \frac{1}{T_V - h} \sum_{\tau=T_E}^{T_E+T_V-h} \left(y_{\tau+h} - G^e(\mathcal{X}_\tau, \hat{\beta}_{j,h,\tau}^e(\mathbf{X}_{T_E}, \lambda_t^e)) \right)^2 + \underbrace{\lambda_{1,t}^e \sum_{k=1}^K |\hat{\beta}_{j,h,t,k}^e|}_{L_1 \text{ Penalty}} + \underbrace{\lambda_{2,t}^e \sum_{k=1}^K (\hat{\beta}_{j,h,t,k}^e)^2}_{L_2 \text{ Penalty}} \right\}$$

where $\hat{\beta}_{j,h,\tau}^e(\cdot)$ for $e \in \{\text{EN}, \text{LSTM}\}$ is the time τ estimate of $\beta_{j,h}^e$ given λ_t^e and data through time τ in a training sample of size T_E . Denote the combined final estimator $\hat{\beta}_{h,t}^e(\mathbf{X}_{\hat{T}_E}, \hat{\lambda}_t^e)$, where the regularization parameter $\hat{\lambda}_t^e$ is estimated using cross-validation dynamically over time. Note that the algorithm also asks the machine to dynamically choose both the optimal training window $\hat{T}_E$ and the optimal validation window $\hat{T}_V$ by minimizing the pseudo-out-of-sample MSE.

The estimation of $\hat{\beta}_{h,t}^e(\mathbf{X}_{\hat{T}_E}, \hat{\lambda}_t^e)$ is repeated sequentially in rolling subsamples, with parameters estimated from information known at time t . Note that the time t subscripts of $\hat{\beta}_{h,t}^e$ and $\hat{\lambda}_t^e$ denote one in a sequence of time-invariant parameter estimates obtained from rolling subsamples, rather than estimates that vary over time within a sample. Likewise, I denote the time t machine belief about y_{t+h} as $\mathbb{E}_t^e[y_{t+h}]$, defined by

$$\mathbb{E}_t^e[y_{t+h}] \equiv G^e(\mathcal{X}_t, \hat{\beta}_{h,t}^e(\mathbf{X}_{\hat{T}_E}, \hat{\lambda}_t^e))$$

Finally, the machine MSE is computed by averaging across the sequence of squared forecast errors in the true out-of-sample forecasts for periods $t = (\hat{T} + h), \dots, T$ where T is the last period of my sample. The true out-of-sample forecasts used for neither estimation nor tuning is the *testing subsample* used to evaluate the model's predictive performance.

On rare occasions, one or more of the explanatory variables used in the machine forecast specification assumes a value that is order of magnitudes different from its historical value. This is usually indicative of a measurement problem in the

raw data. I therefore program the machine to detect in real-time whether its forecast is an extreme outlier, and in that case to discard the forecast replacing it with the historical mean. Specifically, at each t , the machine forecast $\mathbb{E}_t^e[y_{t+h}]$ is set to be the historical mean calculated up to time t whenever the former is five or more standard deviations above its own rolling mean over the most recent 20 quarters.

I include the contemporaneous survey forecasts $\mathbb{F}_t[y_{t+h}]$ for the median respondent only for inflation and GDP forecasts, following Bianchi et al. (2022). This procedure allows the machine to capture intangible information due to judgement or private signals. Specifically, for these forecasts of inflation and GDP growth, I consider the following machine learning empirical specification for forecasting y_{t+h} given information at time t , to be benchmarked against the time t survey forecast of respondent-type X , where this type is the median here:

$$y_{t+h} = G_{jh}^e(\mathbf{Z}_t) + \gamma_{jhM} \mathbb{F}_t[y_{t+h}] + \epsilon_{t+h}, \quad h \geq 1$$

where γ_{jhM} is a parameter to be estimated, and where $G_{jhM}(\mathbf{Z}_t)$ represents a ML estimator as function of big data. Note that the intercept α_{jh} from Bianchi et al. (2022) gets absorbed into the $G_{jh}^e(\mathbf{Z}_t)$ in LSTM via the outermost bias term.

OD.1.1 Elastic Net (EN)

I use the Elastic Net (EN) estimator, which combines Least Absolute Shrinkage and Selection Operator (LASSO) and ridge type penalties. The model can be written as:

$$y_{t+h} = \mathcal{X}'_{tj} \boldsymbol{\beta}_{j,h}^{\text{EN}} + \epsilon_{t+h}$$

where $\mathcal{X}_t = (1, \mathcal{X}_{1t}, \dots, \mathcal{X}_{Kt})'$ include the independent variable observations $(\mathbb{F}_t[y_{t+h}], \mathcal{Z}_{j,t})$ into a vector with "1" and $\boldsymbol{\beta}_{j,h}^{\text{EN}} = (\alpha_{j,h}, \beta_{j,h\mathbb{F}}, \text{vec}(\mathbf{B}_{j,h\mathcal{Z}}))' \equiv (\beta_0, \beta_1, \dots, \beta_K)'$ collects all the coefficients.

It is customary to standardize the elements of $\mathcal{X}_t$ such that sample means are zero and sample standard deviations are unity. The coefficient estimates are then put back in their original scale by multiplying the slope coefficients by their respective standard deviations, and adding back the mean (scaled by slope coefficient over standard deviation.) The EN estimator incorporates both an L_1 and L_2 penalty:

$$\hat{\boldsymbol{\beta}}_{j,h}^{\text{EN}} = \underset{\beta_0, \beta_1, \dots, \beta_K}{\text{argmin}} \frac{1}{T_E} \sum_{\tau=1}^{T_E} \left(y_{\tau+h} - \mathcal{X}'_{\tau j} \boldsymbol{\beta}_{j,h} \right)^2 + \underbrace{\lambda_1 \sum_{k=1}^K |\beta_{j,h,k}|}_{\text{LASSO}} + \underbrace{\lambda_2 \sum_{k=1}^K (\beta_{j,h,k})^2}_{\text{ridge}}$$

By minimizing the MSE over the training samples, I choose the optimal λ_1 and λ_2 values simultaneously.

In the implementation, the EN estimator is sometimes used as an input into the algorithm using the LSTM estimator. Specifically, I ensure that the machine forecast can only differ from the relevant benchmark if it demonstrably improves the pseudo out-of-sample prediction in the training samples *prior* to making a true out-of-sample forecast. Otherwise, the machine is replaced by the benchmark calculated up to time t . In some cases the benchmark is a survey forecast, in others it could be a historical mean value for the variable. However, for the implementation using LSTM, I also use the EN forecast as a benchmark.

OD.1.2 Long Short-Term Memory (LSTM) Network

An LSTM network is a type of Recurrent Neural Network (RNN), which are neural networks used to learn about sequential data such as time series or natural language. In particular, LSTM networks can learn long-term dependencies between across time periods by introducing hidden layers and memory cells to control the flow of information over longer

time periods. The general case of the LSTM network with up to N hidden layers is defined as

$$\begin{aligned}
\underbrace{G^{\text{LSTM}}(\mathcal{X}_t, \beta_{j,h}^{\text{LSTM}})}_{1 \times 1} &= \underbrace{W^{(y h^N)}}_{1 \times D_{h^N}} \underbrace{h_t^N}_{D_{h^N} \times 1} + \underbrace{b_y}_{1 \times 1} & (\text{Output layer}) \\
\underbrace{h_t^n}_{D_{h^n} \times 1} &= \underbrace{o_t^n}_{D_{h^n} \times 1} \odot \tanh(\underbrace{c_t^n}_{D_{h^n} \times 1}) & (\text{Hidden layer}) \\
\underbrace{c_t^n}_{D_{h^n} \times 1} &= \underbrace{f_t^n}_{D_{h^n} \times 1} \odot \underbrace{c_{t-1}^n}_{D_{h^n} \times 1} + \underbrace{i_t^n}_{D_{h^n} \times 1} \odot \underbrace{\tilde{c}_t^n}_{D_{h^n} \times 1} & (\text{Final memory}) \\
\underbrace{\tilde{c}_t^n}_{D_{h^n} \times 1} &= \tanh(\underbrace{W^{(c^n h^{n-1})}}_{D_{h^n} \times D_{h^{n-1}}} \underbrace{h_{t-1}^{n-1}}_{D_{h^{n-1}} \times 1} + \underbrace{W^{(c^n h^n)}}_{D_{h^n} \times D_{h^n}} \underbrace{h_{t-1}^n}_{D_{h^n} \times 1} + \underbrace{b_{c^n}}_{D_{h^n} \times 1}) & (\text{New memory}) \\
\underbrace{f_t^n}_{D_{h^n} \times 1} &= \sigma(\underbrace{W^{(f^n h^{n-1})}}_{D_{h^n} \times D_{h^{n-1}}} \underbrace{h_{t-1}^{n-1}}_{D_{h^{n-1}} \times 1} + \underbrace{W^{(f^n h^n)}}_{D_{h^n} \times D_{h^n}} \underbrace{h_{t-1}^n}_{D_{h^n} \times 1} + \underbrace{b_{f^n}}_{D_{h^n} \times 1}) & (\text{Forget gate}) \\
\underbrace{i_t^n}_{D_{h^n} \times 1} &= \sigma(\underbrace{W^{(i^n h^{n-1})}}_{D_{h^n} \times D_{h^{n-1}}} \underbrace{h_{t-1}^{n-1}}_{D_{h^{n-1}} \times 1} + \underbrace{W^{(i^n h^n)}}_{D_{h^n} \times D_{h^n}} \underbrace{h_{t-1}^n}_{D_{h^n} \times 1} + \underbrace{b_{i^n}}_{D_{h^n} \times 1}) & (\text{Input gate}) \\
\underbrace{o_t^n}_{D_{h^n} \times 1} &= \sigma(\underbrace{W^{(o^n h^{n-1})}}_{D_{h^n} \times D_{h^{n-1}}} \underbrace{h_{t-1}^{n-1}}_{D_{h^{n-1}} \times 1} + \underbrace{W^{(o^n h^n)}}_{D_{h^n} \times D_{h^n}} \underbrace{h_{t-1}^n}_{D_{h^n} \times 1} + \underbrace{b_{o^n}}_{D_{h^n} \times 1}) & (\text{Output gate})
\end{aligned}$$

where $n = 1, \dots, N$ indexes each hidden layer. $h_t^n \in \mathbb{R}^{D_{h^n}}$ is the n -th *hidden layer*, where D_{h^n} is the number of *neurons* or *nodes* in the hidden layer. The 0-th layer is defined as the input data: $h_t^0 \equiv \mathcal{X}_t$. The memory cell c_t^n allows the LSTM network to retain information over longer time periods. The output gate o_t^n controls the extent to which the memory cell c_t^n maps to the hidden layer h_t^n . The forget gate f_t^n controls the flow of information carried over from the final memory in the previous timestep c_{t-1}^n . The input gate i_t^n controls the flow of information from the new memory cell $\tilde{c}_t^n$. The initial states for the hidden layers (h_0^n) $_{n=1}^N$ and memory cells (c_0^n) $_{n=1}^N$ are set to zeros. $\sigma(\cdot)$ and $\tanh(\cdot)$ are *activation functions* that introduce non-linearities in the LSTM network, applied elementwise. $\sigma: \mathbb{R} \rightarrow \mathbb{R}$ is the sigmoid function: $\sigma(x) = (1 + e^{-x})^{-1}$. $\tanh: \mathbb{R} \rightarrow \mathbb{R}$ is the hyperbolic tangent function: $\tanh(x) = \frac{e^{2x}-1}{e^{2x}+1}$. The $\odot$ operator refers to elementwise multiplication. $\beta_{j,h}^{\text{LSTM}} \equiv (((\text{vec}(W^{(g^n h^{n-1})})', \text{vec}(W^{(g^n h^n)})', b'_{g^n})_{g \in \{c, f, i, o\}})_{n=1}^N, \text{vec}(W^{(y h^N)})', b_y)'$ are parameters to be estimated. I will refer to parameters indexed with W as *weights*; parameters indexed with b are *biases*. I estimate the parameters $\beta_{j,h}^{\text{LSTM}}$ for the LSTM network using Stochastic Gradient Decent (SGD), which is an iterative algorithm for minimizing the loss function and proceeds as follows:

1. *Initialization.* Fix a random seed R and draw a starting value of the parameters $\beta_{j,h}^{(0)}$ randomly, where the superscript (0) in parentheses indexes the iteration for an estimate of $\beta_{j,h}^{\text{LSTM}}$.

- (a) Initialize input weights $W^{(g^n h^{n-1})} \in \mathbb{R}^{D_{h^n} \times D_{h^{n-1}}}$ for $g \in \{c, f, i, o\}$ using the *Glorot* initializer. Draw from a uniform distribution with zero mean and a variance that depends on the dimensions of the matrix:

$$W_{ij}^{(g^n h^{n-1})} \stackrel{iid}{\sim} U \left[-\sqrt{\frac{6}{D_{h^n} + D_{h^{n-1}}}}, \sqrt{\frac{6}{D_{h^n} + D_{h^{n-1}}}} \right]$$

for each $i = 1, \dots, D_{h^n}$ and $j = 1, \dots, D_{h^{n-1}}$.

- (b) Initialize the recurrent weights $W^{(g^n h^n)} \in \mathbb{R}^{D_{h^n} \times D_{h^n}}$ for $g \in \{c, f, i, o\}$ using the *Orthogonal* initializer. Use the orthogonal matrix obtained from the QR decomposition of a $D_{h^n} \times D_{h^n}$ matrix of random numbers drawn from a standard normal distribution.
- (c) Initialize biases $(b_{g^n})_{g \in \{c, f, i, o\}}$, hidden layers h_0^n , and memory cells c_0^n with zeros.

2. *Mini-batches.* Prepare the input data by dividing the training sample into a collection of *mini-batches*.

- (a) Suppose that I have a multi-variate time-series training sample with dimensions (T_E, K) whose time steps t are indexed by $t = 1, \dots, T_E$ and K is the number of predictors. I transform this training sample into a 3-D tensor with dimensions (N_S, M, K) where
 - N_S = Total number of sequences in training sample
 - M = Sequence length, i.e., number of time steps in each sequence
 - K = Input size, i.e., number of predictors in each time step

This can be done by creating overlapping sequences from the time series:

- Sequence 1 contains time steps $1, \dots, M$
 - Sequence 2 contains time steps $2, \dots, M + 1$
 - Sequence 3 contains time steps $3, \dots, M + 2$
 - $\dots$
 - Sequence $T_E - M$ contains time steps $T_E - M, \dots, T_E - 1$
 - Sequence $N_S = T_E - M + 1$ contains time steps $T_E - M + 1, \dots, T_E$
- (b) Randomly shuffle the N_S sequences by randomly sampling a permutation without replacement.
- (c) Partition the N_S shuffled sequences into $\lceil N_S/N_B \rceil$ mini-batches. I partition the N_S sequences in the training sample $((N_S, M, K)$ tensor) into a list of $\lceil N_S/N_B \rceil$ mini-batches. A mini-batch is a (N_B, M, K) -dimensional tensor containing N_B out of N_S randomly shuffled sequences. When N_S/N_B is not a whole number, $\lfloor N_S/N_B \rfloor$ of the mini-batches will be 3-D tensors with dimensions (N_B, M, K) . One batch will contain leftover sequences and will have dimensions $(N_S \% N_B, M, K)$ where $\%$ is the modulus operator. Let $B^{(1)}, \dots, B^{\lceil N_S/N_B \rceil}$ denote the list of mini-batches.
- N_S = Total number of sequences in training sample
 - N_B = Mini-batch size, i.e., number of sequences in each partition.
 - M = Sequence length, i.e., number of time steps in each sequence
 - K = Input size, i.e., number of predictors in each time step
3. Repeat until the stopping condition is satisfied ($k = 1, 2, 3, \dots$):

- (a) *Dropout*. Apply dropout to the mini-batch. To obtain the n -th hidden layer under dropout, multiply the current value of the $n-1$ -th hidden layer h_t^{n-1} and the lagged value of the n -th hidden layer h_{t-1}^n with binary masks $r_{t, h_t^{n-1}}^{(k)} \in \mathbb{R}^{D_{h^{n-1}}}$ and $r_{t, h_{t-1}^n}^{(k)} \in \mathbb{R}^{D_{h^n}}$, respectively:

$$\begin{aligned} \underbrace{\bar{h}_t^{n-1}}_{D_{h^{n-1}} \times 1} &= \underbrace{r_{t, h_t^{n-1}}^{(k)}}_{D_{h^{n-1}} \times 1} \odot \underbrace{h_t^{n-1}}_{D_{h^{n-1}} \times 1}, \quad r_{t, h_t^{n-1}, i}^{(k)} \stackrel{iid}{\sim} \text{Bernoulli}(p_{h_t^{n-1}, i}), \quad i = 1, \dots, D_{h^{n-1}} \\ \underbrace{\bar{h}_{t-1}^n}_{D_{h^n} \times 1} &= \underbrace{r_{t, h_{t-1}^n}^{(k)}}_{D_{h^n} \times 1} \odot \underbrace{h_{t-1}^n}_{D_{h^n} \times 1}, \quad r_{t, h_{t-1}^n, i}^{(k)} \stackrel{iid}{\sim} \text{Bernoulli}(p_{h_{t-1}^n, i}), \quad i = 1, \dots, D_{h^n} \end{aligned}$$

where $t \in B^{(k)}$ and $n = 1, \dots, N$ indexes the hidden layer and it is understood that the 0-th layer is the input vector $h_t^0 \equiv \mathcal{X}_t$. $p_{h_t^{n-1}, i}, p_{h_{t-1}^n, i} \in [0, 1]$ is the probability that time t nodes in the $n-1$ -th hidden layer and time $t-1$ nodes in the n -th hidden layer are retained, respectively.

- (b) *Stochastic Gradient*. Average the gradient over observations in the mini-batch

$$\nabla L(\beta_{j,h}^{(k-1)}, \mathbf{X}_{B^{(k)}}, \boldsymbol{\lambda}^{\text{LSTM}}) = \frac{1}{M} \sum_{t \in B^{(k)}} \nabla L(\beta_{j,h}^{(k-1)}, \mathbf{X}_t, \boldsymbol{\lambda}^{\text{LSTM}})$$

where $\nabla L(\beta_{j,h}^{(k-1)}, \mathbf{X}_t, \boldsymbol{\lambda}^{\text{LSTM}})$ is the gradient of the loss function with respect to the parameters $\beta_{j,h}^{(k-1)}$, evaluated at the time t observation $\mathbf{X}_t = (y_{t+h}, \hat{\mathcal{X}}_t)'$ after applying dropout.

- (c) *Learning rate shrinkage*. Update the parameters to $\beta_{j,h}^{(k)}$ using the Adaptive Moment Estimation (Adam) algorithm. The method uses the first and second moments of the gradients to shrink the overall learning rate to zero as the gradient approaches zero.

$$\beta_{j,h}^{(k)} = \beta_{j,h}^{(k-1)} - \gamma \frac{m^{(k)}}{\sqrt{v^{(k)} + \epsilon}}$$

where $m^{(k)}$ and $v^{(k)}$ are weighted averages of first two moments of past gradients:

$$\begin{aligned} m^{(k)} &= \frac{1}{1 - \pi_1^k} (\pi_1 m^{(k-1)} + (1 - \pi_1) \nabla L(\beta_{j,h}^{(k-1)}, \mathbf{X}_{B^{(k)}}, \boldsymbol{\lambda}^{\text{LSTM}})) \\ v^{(k)} &= \frac{1}{1 - \pi_2^k} (\pi_2 v^{(k-1)} + (1 - \pi_2) \nabla L(\beta_{j,h}^{(k-1)}, \mathbf{X}_{B^{(k)}}, \boldsymbol{\lambda}^{\text{LSTM}})^2) \end{aligned}$$

π^k denotes the k -th power of $\pi \in (0, 1)$, and $/$, $\sqrt{\cdot}$, and $(\cdot)^2$ are applied elementwise. The default values of the hyperparameters are $m^{(0)} = v^{(0)} = 0$ (initial moment vectors), $\gamma = 0.001$ (initial learning rate), $(\pi_1, \pi_2) = (0.9, 0.999)$ (decay rates), and $\epsilon = 10^{-7}$ (prevent zero denominators).

(d) *Stopping Criteria.* Stop iterating and return $\beta_{j,h}^{(k)}$ if one of the following holds:

- *Early stopping.* At each iteration, use the updated $\beta_{j,h}^{(k)}$ to calculate the loss from the validation sample. Stop when the validation loss has not improved for S steps, where S is a “patience” hyperparameter. By updating the parameters for fewer iterations, early stopping shrinks the final parameters $\beta_{j,h}$ towards the initial guess $\beta_{j,h}^{(0)}$, and at a lower computational cost than ℓ_2 regularization.
- *Maximum number of epochs.* Stop if the number of iterations reaches the maximum number of epochs E . An epoch happens when the full set of the training sample has been used to update the parameters. If the training sample has T_E observations and each mini-batch has M observations, then each epoch would contain $\lceil T_E/M \rceil$ iterations (after rounding up as needed). So the maximum number of iterations is bounded by $E \times \lceil T_E/M \rceil$.

4. *Ensemble forecasts.* Repeat steps 1. and 2. over different random seeds R and save each of the estimated parameters $\hat{\beta}_{j,h,T_E}^{LSTM}(\mathbf{X}_{T_E}, \boldsymbol{\lambda}^{LSTM}, R)$. Then construct out-of-sample forecasts using the top 10 out of 20 starting values with the best performance in the validation sample. Ensemble can be considered as a regularization method because it aims to guard against overfitting by shrinking the forecasts toward the average across different random seeds. The random seed affects the random draws of the parameter’s initial starting value $\beta_{j,h}^{(0)}$, the sequences selected in each mini-batch $B^{(k)}$, and the dropout mask $r_t^{(k)}$.

Hyperparameters Let $\boldsymbol{\lambda}^{LSTM} \equiv [\lambda_1, \lambda_2, \gamma, \pi_1, \pi_2, p, N, (D_{h^n})_{n=1}^N, M, E, S]'$ collect all the hyper-parameters that control the LSTM network’s complexity and prevent the model from overfitting the data. The number of hidden layers N and the number of neurons $D_{h1}, \dots, D_{hN}$ in each hidden layer are hyper-parameters that characterize the network’s architecture. To choose the number of neurons in each layer, I apply a geometric pyramid rule where the dimension of each additional hidden layer is half that of the previous hidden layer. I select the best LSTM architecture iteratively by minimizing the pseudo out-of-sample mean-squared error from rolling forecasts over the validation sample. Table OA.3 reports the hyper-parameters for the LSTM network and its estimation. Hyper-parameters reported as a range or a set of values are cross-validated. The hyper-parameters are estimated by minimizing the mean-square loss over pseudo out-of-sample forecast errors generated from rolling regressions through the validation sample. The pseudo out-of-sample forecasts are ensemble averages implied by parameters based on different random seeds R .

Adaptive Architecture Selection I allow the LSTM architecture to evolve over time using a simple, adaptive updating procedure. At each period in the testing sample, the machine selects the architecture (number of hidden layers and neurons per layer) that minimized out-of-sample forecast errors in the preceding period. The candidate architectures considered span various combinations of hidden layers and neurons per layer, as listed in Table OA.3. The architecture is updated quarterly by using the forecast performance from the most recent quarter. This approach allows the machine to adjust its specification over time based on evolving patterns in the data, while avoiding look-ahead bias or overfitting to future outcomes.

OD.2 Data Inputs for Machine Learning Algorithm

OD.2.1 Macro Data Surprises

These data are used as inputs into the machine learning forecasts. I obtain median forecasts for GDP growth (Q/Q percentage change), core CPI (Month/Month change), unemployment rate (percentage point), and nonfarm payroll (month/month change) from the Money Market Service Survey. The median market survey forecasts are compiled and published by the Money Market Services (MMS) the Friday before each release. I apply the approach used in Bauer and Swanson (2023) and define macroeconomic data surprise as the actual value of the data release minus the median expectation from MMS on the Friday immediately prior to that data release. The GDP growth forecasts are available quarterly from 1990Q1 to 2023Q4. The core CPI forecast is available monthly from July 1989 to December 2023. The median forecasts for the unemployment rate and nonfarm payrolls are available monthly from January 1980 to December 2023, and January 1985 to December 2023, respectively. All survey forecasts were downloaded from Haver Analytics on December 17, 2022 and the Bloomberg Terminal on July 15, 2025. To pin down the timing of when the news was actually released I follow the published tables of releases from the Bureau of Labor Statistics (BLS), discussed below.

The macro news events are indexed by their date and time of the data release, while the machine learning algorithm is adapted to quarterly sampling frequencies. When including the macro data surprises as additional predictors for the machine forecast, I time-aggregate the macro data surprises to a quarterly frequency by taking the sum of the surprises across data releases that occurred before the response deadline set for the machine. For example, if the response deadline is set to the first day of the middle month of each quarter (e.g., February 1st), I take the sum of the surprises from data releases up to the day before the deadline, the last day of the first month of each quarter (e.g., January 31st).

Table OA.3: Candidate hyper-parameters for the machine learning forecast

Variable	Earnings Growth	Stock Returns
Horizon (Years)	1,2,3,4,5	1,2,3,4,5
(a) Elastic Net		
L_1 penalty λ_1	$[10^{-2}, 10^1]$	$[10^{-6}, 10^{-2}]$
L_2 penalty λ_2	$[10^{-2}, 10^1]$	$[10^{-6}, 10^{-2}]$
Training window T_E	4, 6, 8, 10	5, 7
Validation window T_V	4, 6, 8, 10	5, 7, 20
(b) Long Short-Term Memory Network		
L_1 penalty λ_1	$[10^{-6}, 10^{-2}]$	$[10^{-6}, 10^{-2}]$
L_2 penalty λ_2	$[10^{-6}, 10^{-2}]$	$[10^{-6}, 10^{-2}]$
Learning rate γ	0.001	0.001
Gradient decay π_1, π_2	0.9, 0.999	0.9, 0.999
Dropout input p_x	0.5	0.5
Dropout recurrent p_h	0.5	0.5
Hidden layers N	1, 3, 5	1, 3, 5
Neurons per layer	16, 32, 64	4, 8, 16
Mini-batch size M	4	4
Max epochs E	10,000	10,000
Early stopping S	20	20
Random seeds R	1, ..., 20	1, ..., 20
Training window T_E	4, 8, 12	5, 7
Validation window T_V	4, 8, 12	5, 7, 20

Notes: This table reports the hyperparameters considered in the machine learning algorithm for each estimator.

OD.2.2 FOMC Surprises

FOMC surprises are defined as the changes in the current-month, 1, 2, 6, 12, and 24 month-ahead federal funds futures (FFF) contract rate and changes in the 1, 2, 4, and 8 quarter-ahead Eurodollar (ED) futures contract rate, from 10 minutes before to 20 minutes after each U.S. Federal Reserve Federal Open Market Committee (FOMC) announcement. The data on FFF and ED were downloaded on July 15, 2025. When benchmarking against a survey, I use the last FOMC meeting before the survey deadline to compute surprises. For surveys that do not have a clear deadline, I compute surprises using the last FOMC in the first month of the quarter. When benchmarking against moving average, I use the last FOMC meeting before the end of the first month in each quarter to compute surprises.

When including the FOMC surprises as additional predictors for the machine forecast, I time-aggregate the FOMC surprises to a quarterly frequency by taking the sum of the surprises across FOMC announcements that occurred before the response deadline set for the machine. For example, if the response deadline is set to the first day of the middle month of each quarter (e.g., February 1st), I take the sum of the surprises from FOMC announcements up to the day before the deadline, the last day of the first month of each quarter (e.g., January 31st).

OD.2.3 S&P 500 Jumps

As a measure of the market's reaction to news shocks, I use the jump in the S&P 500 pre- and post- a 30-minute window around major news events. The events in my analysis include (i) 1,482 macroeconomic data releases for U.S. GDP, Consumer Price Index (CPI), unemployment, and payroll data spanning 1980:01-2023:12, (ii) 16 corporate earnings announcement days spanning 1999:03-2020:05, and (iii) 219 Federal Open Market Committee (FOMC) press releases from the Fed spanning 1994:02-2023:12. The corporate earnings news events are from Baker et al. (2019) who conduct textual analyses of *Wall Street Journal* articles to identify days in which there were large jumps in the aggregate stock market that could be attributed to corporate earnings news with high confidence. The jump in the S&P 500 for a given event is defined as $j_\tau = p_{\tau+\delta_{post}} - p_{\tau-\delta_{pre}}$, where τ indexes the time of an event and $p_\tau = \log(P_\tau)$ is the log S&P 500 index. δ_{pre} and δ_{post} denote the pre and post event windows, which is 10 minutes before and 20 minutes after the event, respectively. I obtain data on P_τ using tick-by-tick data on the S&P 500 index from tickdata.com. The series was purchased and downloaded on July 15, 2025 from <https://www.tickdata.com/>. I create the minutely data using the close price within each minute. I supplement the S&P 500 index using S&P500 E-mini futures for events that occur in off-market hours. I use the current-quarter contract futures. I purchased the S&P 500 E-mini futures from CME group on July 15, 2025 at <https://datamine.cmegroup.com/>. My sample spans 1/2/1986 to 12/31/2023.

For each event, I separate out the events for which the S&P 500 increased over the window ($j_\tau^{(+)} \geq 0$) and those for which the market decreased ($j_\tau^{(-)} \leq 0$). I aggregate the event-level jumps to monthly time series by summing over all the relevant events within the month, where the events are partitioned into two groups based on the sign of the jump: $J_t^{(+)} = \sum_{\tau \in x(t)} j_\tau^{(+)}$, $J_t^{(-)} = \sum_{\tau \in x(t)} j_\tau^{(-)}$, where t indexes the month and $x(t)$ is the set of all events that occurred within month t . The procedure results in two monthly variables, $J_t^{(+)}$ and $J_t^{(-)}$, which capture total market reaction to news

events in either direction during the quarter. The series spans the period 1994:02 to 2023:12. Separating out the events based on the sign of the jump allows me to capture any differential effects on return predictability based on whether the market perceived the news as good or bad. The partition also allows me to accurately capture the total extent of over- or underreaction. Otherwise, mixing all the events would only capture the net effect of the jumps and bias the market reaction towards zero.

When used as additional predictors for the machine forecast, the jumps need to be converted to quarterly time series because the machine learning algorithm is adapted to a quarterly sampling frequency. The set of events in $x(t)$ is chosen so that the machine only sees the news events that would have been available to the real-time firm. When combining the events within a quarter, I impose the response deadline used to produce the machine forecast. For example, if the response deadline is set to the first day of the middle month of each quarter (e.g., February 1st), I use the jumps from the events up to the day before the deadline, the last day of the first month of each quarter (e.g., January 31st).

OD.2.4 Real-Time Macro Data

This section gives details on the real time macro data inputs used in the machine learning forecasts. A subset of these series are used in the structural estimation. At each forecast date in the sample, I construct a dataset of macro variables that could have been observed on or before the day of the survey deadline. I use the Philadelphia Fed's Real-Time Data Set to obtain vintages of macro variables. The real-time data sets are available at <https://www.philadelphiafed.org/research-and-data/real-time-center/real-time-data/data-files>. These vintages capture changes to historical data due to periodic revisions made by government statistical agencies. The vintages for a particular series can be available at the monthly and/or quarterly frequencies, and the series have monthly and/or quarterly observations. In cases where a variable has both frequencies available for its vintages and/or its observations, I choose one format of the variable. For instance, nominal personal consumption expenditures on goods is quarterly data with both monthly and quarterly vintages available; in this case, I use the version with monthly vintages.

Table OA.4 gives the complete list of real-time macro variables. Included in the table is the first available vintages for each variable that has multiple vintages. I do not include the last vintage because most variables have vintages through the present. For variables BASEBASAQVMD, NBRBASAQVMD, NBRECBASAQVMD, and TRBASAQVMD, the last available vintage is 2013Q2. Table OA.4 also lists the transformation applied to each variable to make them stationary before generating factors. Let $X_{i,t}$ denote variable i at time t after the transformation, and let $X_{i,t}^A$ be the untransformed series. Let $\Delta = (1 - L)$ with $LX_{i,t} = X_{i,t-1}$. There are seven possible transformations with the following codes:

- 1 Code lv : $X_{i,t} = X_{i,t}^A$
- 2 Code Δlv : $X_{i,t} = X_{i,t}^A - X_{i,t-1}^A$
- 3 Code $\Delta^2 lv$: $X_{i,t} = \Delta^2 X_{i,t}^A$
- 4 Code ln : $X_{i,t} = \ln(X_{i,t}^A)$
- 5 Code Δln : $X_{i,t} = \ln(X_{i,t}^A) - \ln(X_{i,t-1}^A)$
- 6 Code $\Delta^2 ln$: $X_{i,t} = \Delta^2 \ln(X_{i,t}^A)$
- 7 Code $\Delta lv/lv$: $X_{i,t} = (X_{i,t}^A - X_{i,t-1}^A)/X_{i,t-1}^A$

Table OA.4: List of Macro Dataset Variables

No.	Short Name	Source	Tran	Description	First Vintage
Group 1: Output and Income					
1	IPMMVMD	Philly Fed	Δln	Ind. production index - Manufacturing	1962M11
2	IPTMVMD	Philly Fed	Δln	Ind. production index - Total	1962M11
3	CUMMVMD	Philly Fed	lv	Capacity utilization - Manufacturing	1979M8
4	CUTMVMD	Philly Fed	lv	Capacity utilization - Total	1983M7
5	NCPROFATMVQD	Philly Fed	Δln	Nom. corp. profits after tax without IVA/CCAdj	1965Q4
6	NCPROFATWMVQD	Philly Fed	Δln	Nom. corp. profits after tax with IVA/CCAdj	1981Q1
7	OPHMVQD	Philly Fed	Δln	Output per hour - Business sector	1998Q4
8	NDPIQVQD	Philly Fed	Δln	Nom. disposable personal income	1965Q4
9	NOUTPUTQVQD	Philly Fed	Δln	Nom. GNP/GDP	1965Q4
10	NPIQVQD	Philly Fed	Δln	Nom. personal income	1965Q4
11	NPSAVQVQD	Philly Fed	Δlv	Nom. personal saving	1965Q4
12	OLIQVQD	Philly Fed	Δln	Other labor income	1965Q4
13	PINTIQVQD	Philly Fed	Δln	Personal interest income	1965Q4
14	PINTPAIDQVQD	Philly Fed	Δln	Interest paid by consumers	1965Q4
15	PROPIQVQD	Philly Fed	Δln	Proprietors' income	1965Q4
16	PTAXQVQD	Philly Fed	Δln	Personal tax and nontax payments	1965Q4
17	RATESAVQVQD	Philly Fed	Δlv	Personal saving rate	1965Q4
18	RENTIQVQD	Philly Fed	Δlv	Rental income of persons	1965Q4
19	ROUTPUTQVQD	Philly Fed	Δln	Real GNP/GDP	1965Q4
20	SSCONTRIBQVQD	Philly Fed	Δln	Personal contributions for social insurance	1965Q4

No.	Short Name	Source	Tran	Description	First Vintage
21	TRANPFQVQD	Philly Fed	Δln	Personal transfer payments to foreigners	1965Q4
22	TRANRQVQD	Philly Fed	Δln	Transfer payments	1965Q4
23	CUUR0000SA0E	BLS	$\Delta^2 ln$	Energy in U.S. city avg., all urban consumers, not seasonally adj	
Group 2: Employment					
24	EMPLOYMVMD	Philly Fed	Δln	Nonfarm payroll	1946M12
25	HMVMD	Philly Fed	lv	Aggregate weekly hours - Total	1971M9
26	HGMVMD	Philly Fed	lv	Agg. weekly hours - Goods-producing	1971M9
27	HSMVMD	Philly Fed	lv	Agg. weekly hours - Service-producing	1971M9
28	LFCMVMD	Philly Fed	Δln	Civilian labor force	1998M11
29	LFPARTMVMD	Philly Fed	lv	Civilian participation rate	1998M11
30	POPMVMD	Philly Fed	Δln	Civilian noninstitutional population	1998M11
31	ULCMVQD	Philly Fed	Δln	Unit labor costs - Business sector	1998Q4
32	RUCQVMD	Philly Fed	Δlv	Unemployment rate	1965Q4
33	WSDQVQD	Philly Fed	Δln	Wage and salary disbursements	1965Q4
Group 3: Orders, Investment, Housing					
34	HSTARTSMVMD	Philly Fed	Δln	Housing starts	1968M2
35	RINVBFMVQD	Philly Fed	Δln	Real gross private domestic inv. - Nonresidential	1965Q4
36	RINVCHIMVQD	Philly Fed	Δlv	Real gross private domestic inv. - Change in private inventories	1965Q4
37	RINVRESIDMVQD	Philly Fed	Δln	Real gross private domestic inv. - Residential	1965Q4
38	CASESHILLER	S&P	Δln	Case-Shiller US National Home Price index/CPI	1987M1
Group 4: Consumption					
39	NCONGMMVMD	Philly Fed	Δln	Nom. personal cons. exp. - Goods	2009M8
40	NCONHHMMVMD	Philly Fed	Δln	Nom. hh. cons. exp.	2009M8
41	NCONSHHMMVMD	Philly Fed	Δln	Nom. hh. cons. exp. - Services	2009M8
42	NCONSNPMMVMD	Philly Fed	Δln	Nom. final cons. exp. of NPISH	2009M8
43	RCONDMMVMD	Philly Fed	Δln	Real personal cons. exp. - Durables	1998M11
44	RCONGMMVMD	Philly Fed	Δln	Real personal cons. exp. - Goods	2009M8
45	RCONHHMMVMD	Philly Fed	Δln	Real hh. cons. exp.	2009M8
46	RCONMMVMD	Philly Fed	Δln	Real personal cons. exp. - Total	1998M11
47	RCONNDMVMD	Philly Fed	Δln	Real personal cons. exp. - Nondurables	1998M11
48	RCONSHHMMVMD	Philly Fed	Δln	Real hh. cons. exp. - Services	2009M8
49	RCONSMMVMD	Philly Fed	Δln	Real personal cons. exp. - Services	1998M11
50	RCONSNPMMVMD	Philly Fed	Δln	Real final cons. exp. of NPISH	2009M8
51	NCONGMVQD	Philly Fed	Δln	Nom. personal cons. exp. - Goods	2009Q3
52	NCONHHMVQD	Philly Fed	Δln	Nom. hh. cons. exp.	2009Q3
53	NCONSHHMVQD	Philly Fed	Δln	Nom. hh. cons. exp. - Services	2009Q3
54	NCONSNPMVQD	Philly Fed	Δln	Nom. final cons. exp. of NPISH	2009Q3
55	RCONDMVQD	Philly Fed	Δln	Real personal cons. exp. - Durable goods	1965Q4
56	RCONGMVQD	Philly Fed	Δln	Real personal cons. exp. - Goods	2009Q3
57	RCONHHMVQD	Philly Fed	Δln	Real hh. cons. exp.	2009Q3
58	RCONMVQD	Philly Fed	Δln	Real personal cons. exp. - Total	1965Q4
59	RCONNDMVQD	Philly Fed	Δln	Real personal cons. exp. - Nondurable goods	1965Q4
60	RCONSHHMVQD	Philly Fed	Δln	Real hh. cons. exp. - Services	2009Q3
61	RCONSMVQD	Philly Fed	Δln	Real personal cons. exp. - Services	1965Q4
62	RCONSNPMVQD	Philly Fed	Δln	Real final cons. exp. of NPISH	2009Q3
63	NCONQVQD	Philly Fed	Δln	Nom. personal cons. exp.	1965Q4
Group 5: Prices					
64	PCONGMMVMD	Philly Fed	$\Delta^2 ln$	Price index for personal cons. exp. - Goods	2009M8
65	PCONHHMMVMD	Philly Fed	$\Delta^2 ln$	Price index for hh. cons. exp.	2009M8
66	PCONSHHMMVMD	Philly Fed	$\Delta^2 ln$	Price index for hh. cons. exp. - Services	2009M8
67	PCONSNPMMVMD	Philly Fed	$\Delta^2 ln$	Price index for final cons. exp. of NPISH	2009M8
68	PCPIMVMD	Philly Fed	$\Delta^2 ln$	Consumer price index	1998M11
69	PCPIXMVMD	Philly Fed	$\Delta^2 ln$	Core consumer price index	1998M11
70	PPPIMVMD	Philly Fed	$\Delta^2 ln$	Producer price index	1998M11
71	PPPIXMVMD	Philly Fed	$\Delta^2 ln$	Core producer price index	1998M11
72	PCONGMVQD	Philly Fed	$\Delta^2 ln$	Price index for personal cons. exp. - Goods	2009Q3
73	PCONHHMVQD	Philly Fed	$\Delta^2 ln$	Price index for hh. cons. exp.	2009Q3
74	PCONSHHMVQD	Philly Fed	$\Delta^2 ln$	Price index for hh. cons. exp. - Services	2009Q3
75	PCONSNPMVQD	Philly Fed	$\Delta^2 ln$	Price index for final cons. exp. of NPISH	2009Q3
76	PCONXMVQD	Philly Fed	$\Delta^2 ln$	Core price index for personal cons. exp.	1996Q1
77	CPIQVMD	Philly Fed	$\Delta^2 ln$	Consumer price index	1994Q3
78	PQVQD	Philly Fed	$\Delta^2 ln$	Price index for GNP/GDP	1965Q4
79	PCONQVQD	Philly Fed	$\Delta^2 ln$	Price index for personal cons. exp.	1965Q4
80	PIMPQVQD	Philly Fed	$\Delta^2 ln$	Price index for imports of goods and services	1965Q4
Group 6: Trade and Government					
81	REXMVQD	Philly Fed	Δln	Real exports of goods and services	1965Q4
82	RGMVQD	Philly Fed	Δln	Real government cons. and gross inv. - Total	1965Q4
83	RGFMVQD	Philly Fed	Δln	Real government cons. and gross inv. - Federal	1965Q4

No.	Short Name	Source	Tran	Description	First Vintage
84	RGSLMVQD	Philly Fed	Δln	Real government cons. and gross. inv. - State and local	1965Q4
85	RIMPMVQD	Philly Fed	Δln	Real imports of goods and services	1965Q4
86	RNXMVQD	Philly Fed	Δlv	Real net exports of goods and services	1965Q4
Group 7: Money and Credit					
87	BASEBASAQVMD	Philly Fed	$\Delta^2 ln$	Monetary base	1980Q2
88	M1QVMD	Philly Fed	$\Delta^2 ln$	M1 money stock	1965Q4
89	M2QVMD	Philly Fed	$\Delta^2 ln$	M2 money stock	1971Q2
90	NBRBASAQVMD	Philly Fed	$\Delta lv/lv$	Nonborrowed reserves	1967Q3
91	NBRECASAQVMD	Philly Fed	$\Delta lv/lv$	Nonborrowed reserves plus extended credit	1984Q2
92	TRBASAQVMD	Philly Fed	$\Delta^2 ln$	Total reserves	1967Q3
93	DIVQVQD	Philly Fed	Δln	Dividends	1965Q4

OD.2.5 Monthly Financial Data

The 147 financial series in this data set are versions of the financial dataset used in Jurado et al. (2015) and Ludvigson et al. (2021). It consists of a number of indicators measuring the behavior of a broad cross-section of asset returns, as well as some aggregate financial indicators not included in the macro dataset. These data include valuation ratios such as the dividend-price ratio and earnings-price ratio, growth rates of aggregate dividends and prices, default and term spreads, yields on corporate bonds of different ratings grades, yields on Treasuries and yield spreads, and a broad cross-section of industry equity returns. Following Fama and French (1992), returns on 100 portfolios of equities sorted into 10 size and 10 book-to-market categories. The dataset X^f also includes a group of variables I call “risk-factors,” since they have been used in cross-sectional or time-series studies to uncover variation in the market risk-premium. These risk-factors include the three Fama and French (1993) risk factors, namely the excess return on the market MKT_t , the “small-minus-big” (SMB_t) and “high-minus-low” (HML_t) portfolio returns, the momentum factor UMD_t , and the small stock value spread $R15 - R11$.

The raw data used to form factors are always transformed to achieve stationarity. In addition, when forming forecasting factors from the large macro and financial datasets, the raw data (which are in different units) are standardized before performing PCA. When forming common uncertainty from estimates of individual uncertainty, the raw data (which are in this case in the same units) are demeaned, but I do not divide by the observation’s standard deviation before performing PCA. Throughout, the factors are estimated by the method of static principal components (PCA). Specifically, the $T \times r_F$ matrix $\hat{F}_t$ is $\sqrt{T}$ times the r_F eigenvectors corresponding to the r_F largest eigenvalues of the $T \times T$ matrix $xx'/(TN)$ in decreasing order. In large samples (when $\sqrt{T}/N \rightarrow 0$), Bai and Ng (2006) show that the estimates $\hat{F}_t$ can be treated as though they were observed in the subsequent forecasting regression. All returns and spreads are expressed in logs (i.e., the log of the gross return or spread), are displayed in percent (i.e., multiplied by 100), and are annualized by multiplying by 12. That is, if x is the original return or spread, I transform to $1200 \times \log(1 + x/100)$. Federal Reserve data are annualized by default and are therefore not re-annualized. Note that this annualization implies that the annualized standard deviation (volatility) is equal to the data standard deviation divided by $\sqrt{12}$. The data series used in this dataset are listed below by data source. Additional details on data transformations are given below the table.

I convert monthly data to quarterly by using either the beginning-of-quarter or end-of-quarter values. The decision to use beginning-of-quarter or end-of-quarter depends on the survey deadline of a particular forecast date. If the survey deadline is known to be in the middle of the second month of quarter t , then it is conceivable that the forecasters would have information about the first month of quarter t . Therefore, I use the first month of that quarter’s values. Alternatively, a few anomalous observations have unknown survey deadlines (e.g., the SPF deadlines for 1990Q1). In such cases, I allow only information up to quarter $t - 1$ to enter the model. Thus, I use the last month of the previous quarter’s values in these cases. Let $X_{i,t}$ denote variable i observed at time t after, e.g., logarithm and differencing transformation, and let $X_{i,t}^A$ be the actual (untransformed) series. Let $\Delta = (1 - L)$ with $LX_{i,t} = X_{i,t-1}$. There are six possible transformations with the following codes:

- 1 Code lv : $X_{i,t} = X_{i,t}^A$
- 2 Code Δlv : $X_{i,t} = X_{i,t}^A - X_{i,t-1}^A$
- 3 Code $\Delta^2 lv$: $X_{i,t} = \Delta^2 X_{i,t}^A$
- 4 Code ln : $X_{i,t} = \log(X_{i,t}^A)$
- 5 Code Δln : $X_{i,t} = \log(X_{i,t}^A) - \log(X_{i,t-1}^A)$
- 6 Code $\Delta^2 ln$: $X_{i,t} = \Delta^2 \log(X_{i,t}^A)$
- 7 Code $\Delta lv/lv$: $X_{i,t} = \frac{X_{i,t}^A - X_{i,t-1}^A}{X_{i,t-1}^A}$

Table OA.5: List of Financial Dataset Variables

No.	Short Name	Source	Tran	Description
Group 1: Prices, Yields, Dividends				
1	D.log(DIV)	CRSP	$\Delta \ln$	1 log D_t , see additional details below
2	D.log(P)	CRSP	$\Delta \ln$	1 log P_t , see additional details below
3	D.DIVreinvest	CRSP	$\Delta \ln$	1 log $D_t^{re,*}$, see additional details below
4	D.Preinvest	CRSP	$\Delta \ln$	1 log $P_t^{re,*}$, see additional details below
5	d-p	CRSP	$\ln$	log $D_t - P_t$, see additional details below
Group 2: Equity Risk Factors				
6	R15-R11	Kenneth French	lv	(Small, High) minus (Small, Low) sorted on (size, book-to-market)
7	Mkt-RF	Kenneth French	lv	Market excess return
8	SMB	Kenneth French	lv	Small Minus Big, sorted on size
9	HML	Kenneth French	lv	High Minus Low, sorted on book-to-market
10	UMD	Kenneth French	lv	Up Minus Down, sorted on momentum
Group 3: Industries				
11	Agric	Kenneth French	lv	Agric industry portfolio
12	Food	Kenneth French	lv	Food industry portfolio
13	Beer	Kenneth French	lv	Beer industry portfolio
14	Smoke	Kenneth French	lv	Smoke industry portfolio
15	Toys	Kenneth French	lv	Toys industry portfolio
16	Fun	Kenneth French	lv	Fun industry portfolio
17	Books	Kenneth French	lv	Books industry portfolio
18	Hshld	Kenneth French	lv	Hshld industry portfolio
19	Clths	Kenneth French	lv	Clths industry portfolio
20	MedEq	Kenneth French	lv	MedEq industry portfolio
21	Drugs	Kenneth French	lv	Drugs industry portfolio
22	Chems	Kenneth French	lv	Chems industry portfolio
23	Rubbr	Kenneth French	lv	Rubbr industry portfolio
24	Txtls	Kenneth French	lv	Txtls industry portfolio
25	BldMt	Kenneth French	lv	BldMt industry portfolio
26	Cnstr	Kenneth French	lv	Cnstr industry portfolio
27	Steel	Kenneth French	lv	Steel industry portfolio
28	Mach	Kenneth French	lv	Mach industry portfolio
29	ElcEq	Kenneth French	lv	ElcEq industry portfolio
30	Autos	Kenneth French	lv	Autos industry portfolio
31	Aero	Kenneth French	lv	Aero industry portfolio
32	Ships	Kenneth French	lv	Ships industry portfolio
33	Mines	Kenneth French	lv	Mines industry portfolio
34	Coal	Kenneth French	lv	Coal industry portfolio
35	Oil	Kenneth French	lv	Oil industry portfolio
36	Util	Kenneth French	lv	Util industry portfolio
37	Telcm	Kenneth French	lv	Telcm industry portfolio
38	PerSv	Kenneth French	lv	PerSv industry portfolio
39	BusSv	Kenneth French	lv	BusSv industry portfolio
40	Hardw	Kenneth French	lv	Hardw industry portfolio
41	Chips	Kenneth French	lv	Chips industry portfolio
42	LabEq	Kenneth French	lv	LabEq industry portfolio
43	Paper	Kenneth French	lv	Paper industry portfolio
44	Boxes	Kenneth French	lv	Boxes industry portfolio
45	Trans	Kenneth French	lv	Trans industry portfolio
46	Whlsl	Kenneth French	lv	Whlsl industry portfolio
47	Rtail	Kenneth French	lv	Rtail industry portfolio
48	Meals	Kenneth French	lv	Meals industry portfolio
49	Banks	Kenneth French	lv	Banks industry portfolio
50	Insur	Kenneth French	lv	Insur industry portfolio
51	RIEst	Kenneth French	lv	RIEst industry portfolio
52	Fin	Kenneth French	lv	Fin industry portfolio
53	Other	Kenneth French	lv	Other industry portfolio
Group 4: Size/BM				
54	1.2	Kenneth French	lv	(1, 2) portfolio sorted on (size, book-to-market)
55	1.4	Kenneth French	lv	(1, 4) portfolio sorted on (size, book-to-market)
56	1.5	Kenneth French	lv	(1, 5) portfolio sorted on (size, book-to-market)
57	1.6	Kenneth French	lv	(1, 6) portfolio sorted on (size, book-to-market)
58	1.7	Kenneth French	lv	(1, 7) portfolio sorted on (size, book-to-market)
59	1.8	Kenneth French	lv	(1, 8) portfolio sorted on (size, book-to-market)
60	1.9	Kenneth French	lv	(1, 9) portfolio sorted on (size, book-to-market)
61	1.high	Kenneth French	lv	(1, high) portfolio sorted on (size, book-to-market)
62	2.low	Kenneth French	lv	(2, low) portfolio sorted on (size, book-to-market)
63	2.2	Kenneth French	lv	(2, 2) portfolio sorted on (size, book-to-market)
64	2.3	Kenneth French	lv	(2, 3) portfolio sorted on (size, book-to-market)
65	2.4	Kenneth French	lv	(2, 4) portfolio sorted on (size, book-to-market)
66	2.5	Kenneth French	lv	(2, 5) portfolio sorted on (size, book-to-market)
67	2.6	Kenneth French	lv	(2, 6) portfolio sorted on (size, book-to-market)

No.	Short Name	Source	Tran	Description
140	10_low	Kenneth French	lv	(10, low) portfolio sorted on (size, book-to-market)
141	10_2	Kenneth French	lv	(10, 2) portfolio sorted on (size, book-to-market)
142	10_3	Kenneth French	lv	(10, 3) portfolio sorted on (size, book-to-market)
143	10_4	Kenneth French	lv	(10, 4) portfolio sorted on (size, book-to-market)
144	10_5	Kenneth French	lv	(10, 5) portfolio sorted on (size, book-to-market)
145	10_6	Kenneth French	lv	(10, 6) portfolio sorted on (size, book-to-market)
146	10_7	Kenneth French	lv	(10, 7) portfolio sorted on (size, book-to-market)
147	VXO	Fred MD	lv	VXOCLS

CRSP Data Details Value-weighted price and dividend data were obtained from the Center for Research in Security Prices (CRSP, Center for Research in Security Prices (1926–2022)). From the Annual Update data, I obtain the monthly value-weighted return series **vwretd** (with dividends) and **vwretx** (excluding dividends). These series have the interpretations: $VWRET_t = \frac{P_{t+1} + D_{t+1}}{P_t}$, $VWRETX_t = \frac{P_{t+1}}{P_t}$. From these series, a normalized price series P_t can be constructed recursively as: $P_0 = 1$, $P_t = P_{t-1} \times VWRETX_{t-1}$. A dividend series can then be constructed using: $D_t = P_{t-1} \times (VWRET_{t-1} - VWRETX_{t-1})$. In order to remove seasonality of dividend payments from the data, instead of D_t I use the series: $\bar{D}_t = \frac{1}{12} \sum_{j=0}^{11} D_{t-j}$, i.e., the moving average over the entire year. For the price and dividend series under “reinvestment,” I calculate the price under reinvestment, P_t^{re} , as the normalized value of the market portfolio under reinvestment of dividends, using the recursion: $P_0^{re} = 1$, $P_t^{re} = P_{t-1} \times VWRET_{t-1}$. Similarly, I can define dividends under reinvestment, D_t^{re} , as the total dividend payments on this portfolio (the number of “shares” of which have increased over time) using: $D_t^{re} = P_{t-1}^{re} \times (VWRET_{t-1} - VWRETX_{t-1})$. As before, I can remove seasonality by using: $\bar{D}_t^{re} = \frac{1}{12} \sum_{j=0}^{11} D_{t-j}^{re}$. Five data series are constructed from the CRSP data as follows: **D_log(DIV)**: $\Delta \log(\bar{D}_t)$; **D_log(P)**: $\Delta \log(P_t)$; **D_DIVreinvest**: $\Delta \log(\bar{D}_t^{re})$; **D_Preinvest**: $\Delta \log(P_t^{re})$; **d-p**: $\log(\bar{D}_t) - \log(P_t)$.

Kenneth French Data Details The following data are obtained from the data library of Kenneth French’s Dartmouth website (French (1926–2022)):

- Fama/French Factors: From this dataset I obtain the series **RF**, **Mkt-RF**, **SMB**, and **HML**.
- 25 Portfolios Formed on Size and Book-to-Market (5 x 5): From this dataset I obtain the series **R15-R11**, which is the return spread between the (small, high book-to-market) and (small, low book-to-market) portfolios.
- Momentum Factor (Mom): From this dataset I obtain the series **UMD**, which is equal to the momentum factor.
- 49 Industry Portfolios: From this dataset I use all value-weighted series, excluding any series that have missing observations from January 1960 onward. This yields the series **Agric** through **Other**. The omitted series are **Soda**, **Hlth**, **FabPr**, **Guns**, **Gold**, and **Softw**.
- 100 Portfolios Formed on Size and Book-to-Market: From this dataset I use all value-weighted series, excluding any series that have missing observations from January 1960 onward. This yields variables with names **X.Y**, where X denotes the size index (1, 2, ..., 10) and Y denotes the book-to-market index (Low, 2, 3, ..., 8, 9, High). The omitted series are **1_low**, **1_3**, **7_high**, **9_9**, **10_8**, **10_9**, and **10_high**.

VXO Data Details VXO data is obtained from the Monthly Database for Macroeconomic Research (FRED-MD, McCracken (2015–2022)).

OD.2.6 Daily Financial Data

Daily Data and construction of daily factors These data are used in the machine learning forecasts. The daily financial series in this data set are from the daily financial dataset used in Andreou et al. (2013). I create a smaller daily database which is a subset of the large cross-section of 991 daily series in their dataset. My dataset covers five classes of financial assets: (i) the Commodities class; (ii) the Corporate Risk category; (iii) the Equities class; (iv) the Foreign Exchange Rates class and (v) the Government Securities. The dataset includes up to 87 daily predictors in a daily frequency from 23-Oct-1959 to 31-Dec-2023 from the above five categories of financial assets. I remove series with fewer than ten years of data and time periods with no variables observed, which occurs for some series in the early part of the sample. For those years, I have less than 87 series. There are 39 commodity variables which include commodity indices, prices and futures, 16 corporate risk series, 9 equity series which include major US stock market indices and the 500 Implied Volatility, 16 government securities which include the federal funds rate, government treasury bills of securities from three months to ten years, and 7 foreign exchange variables which include the individual foreign exchange rates of major five US trading partners and two effective exchange rate. I choose these daily predictors because they are proposed in the literature as good predictors of economic growth.

I construct daily financial factors in a quarterly frequency in two steps. First, I use these daily financial time series to form factors at a daily frequency. The raw data used to form factors are always transformed to achieve stationarity and

standardized before performing factor estimation (see generic description below). I re-estimate factors at each date in the sample recursively over time using the entire history of data available in real time prior to each out-of-sample forecast. In the second step, I convert these daily financial indicators to quarterly weighted variables to form quarterly factors by selecting an optimal weighting scheme according to the method described below (see the weighting scheme section). The data series used in this dataset are listed below in Table OA.6 by data source. The tables also list the transformation applied to each variable to make them stationary before generating factors. The transformations used to stationarize a time series are the same as those explained in the section “Monthly financial factor data”.

Table OA.6: List of Daily Financial Dataset Variables

No.	Short Name	Source	Tran	Description
Group 1: Commodities				
1	GSIZSPT	Data Stream	$\Delta \ln$	S&P GSCI Zinc Spot - PRICE INDEX
2	GSSBSPT	Data Stream	$\Delta \ln$	S&P GSCI Sugar Spot - PRICE INDEX
3	GSSOSPT	Data Stream	$\Delta \ln$	S&P GSCI Soybeans Spot - PRICE INDEX
4	GSSISPT	Data Stream	$\Delta \ln$	S&P GSCI Silver Spot - PRICE INDEX
5	GSIKSPT	Data Stream	$\Delta \ln$	S&P GSCI Nickel Spot - PRICE INDEX
6	GSLCSPT	Data Stream	$\Delta \ln$	S&P GSCI Live Cattle Spot - PRICE INDEX
7	GSLHSPT	Data Stream	$\Delta \ln$	S&P GSCI Lean Hogs Index Spot - PRICE INDEX
8	GSILSPT	Data Stream	$\Delta \ln$	S&P GSCI Lead Spot - PRICE INDEX
9	GSGCSPT	Data Stream	$\Delta \ln$	S&P GSCI Gold Spot - PRICE INDEX
10	GSCTSPT	Data Stream	$\Delta \ln$	S&P GSCI Cotton Spot - PRICE INDEX
11	GSKCSPT	Data Stream	$\Delta \ln$	S&P GSCI Coffee Spot - PRICE INDEX
12	GSCCSPT	Data Stream	$\Delta \ln$	S&P GSCI Cocoa Index Spot - PRICE INDEX
13	GSIASPT	Data Stream	$\Delta \ln$	S&P GSCI Aluminum Spot - PRICE INDEX
14	SGWTSPT	Data Stream	$\Delta \ln$	S&P GSCI All Wheat Spot - PRICE INDEX
15	EIAEBRT	Data Stream	$\Delta \ln$	Europe Brent Spot FOB U\$/BBL Daily
16	CRUDOIL	Data Stream	$\Delta \ln$	Crude Oil-WTI Spot Cushing U\$/BBL - MID PRICE
17	LTICASH	Data Stream	$\Delta \ln$	LME-Tin 99.85% Cash U\$/MT
18	CWFCS00	Data Stream	$\Delta \ln$	CBT-WHEAT COMPOSITE FUTURES CONT. - SETT. PRICE
19	CCFCS00	Data Stream	$\Delta \ln$	CBT-CORN COMP. CONTINUOUS - SETT. PRICE
20	CSYCS00	Data Stream	$\Delta \ln$	CBT-SOYBEANS COMP. CONT. - SETT. PRICE
21	NCTCS20	Data Stream	$\Delta \ln$	CSCE-COTTON #2 CONT.2ND FUT - SETT. PRICE
22	NSBCS00	Data Stream	$\Delta \ln$	CSCE-SUGAR #11 CONTINUOUS - SETT. PRICE
23	NKCCS00	Data Stream	$\Delta \ln$	CSCE-COFFEE C CONTINUOUS - SETT. PRICE
24	NCCCS00	Data Stream	$\Delta \ln$	CSCE-COCOA CONTINUOUS - SETT. PRICE
25	CZLCS00	Data Stream	$\Delta \ln$	ECBOT-SOYBEAN OIL CONTINUOUS - SETT. PRICE
26	COFC01	Data Stream	$\Delta \ln$	CBT-OATS COMP. TRc1 - SETT. PRICE
27	CLDCS00	Data Stream	$\Delta \ln$	CME-LIVE CATTLE COMP. CONTINUOUS - SETT. PRICE
28	CLGC01	Data Stream	$\Delta \ln$	CME-LEAN HOGS COMP. TRc1 - SETT. PRICE
29	NGCCS00	Data Stream	$\Delta \ln$	CMX-GOLD 100 OZ CONTINUOUS - SETT. PRICE
30	LAH3MTH	Data Stream	$\Delta \ln$	LME-Aluminium 99.7% 3 Months U\$/MT
31	LED3MTH	Data Stream	$\Delta \ln$	LME-Lead 3 Months U\$/MT
32	LNI3MTH	Data Stream	$\Delta \ln$	LME-Nickel 3 Months U\$/MT
33	LTI3MTH	Data Stream	$\Delta \ln$	LME-Tin 99.85% 3 Months U\$/MT
34	PLNYD	www.macrotrends.net	$\Delta \ln$	Platinum Cash Price (U\$ per troy ounce)
35	XPDD	www.macrotrends.net	$\Delta \ln$	Palladium (U\$ per troy ounce)
36	CUS2D	www.macrotrends.net	$\Delta \ln$	Corn Spot Price (U\$/Bushel)
37	SoybOil	www.macrotrends.net	$\Delta \ln$	Soybean Oil Price (U\$/Pound)
38	OATSD	www.macrotrends.net	$\Delta \ln$	Oat Spot Price (US\$/Bushel)
39	WTIOilFut	US EIA	$\Delta \ln$	Light Sweet Crude Oil Futures Price: 1St Expiring Contract Settlement (\$/Bbl)
Group 2: Equities				
40	S&PCOMP	Data Stream	$\Delta \ln$	S&P 500 COMPOSITE - PRICE INDEX
41	ISPCS00	Data Stream	$\Delta \ln$	CME-S&P 500 INDEX CONTINUOUS - SETT. PRICE
42	SP5EIND	Data Stream	$\Delta \ln$	S&P500 ES INDUSTRIALS - PRICE INDEX
43	DJINDUS	Data Stream	$\Delta \ln$	DOW JONES INDUSTRIALS - PRICE INDEX
44	CYMCS00	Data Stream	$\Delta \ln$	CBT-MINI DOW JONES CONTINUOUS - SETT. PRICE
45	NASCOMP	Data Stream	$\Delta \ln$	NASDAQ COMPOSITE - PRICE INDEX
46	NASA100	Data Stream	$\Delta \ln$	NASDAQ 100 - PRICE INDEX
47	CBOEVIX	Data Stream	lv	CBOE SPX VOLATILITY VIX (NEW) - PRICE INDEX
48	S&P500toVIX	Data Stream	$\Delta \ln$	S&P500/VIX
Group 3: Corporate Risk				
49	LIBOR	FRED	Δlv	Overnight London Interbank Offered Rate (%)
50	1MLIBOR	FRED	Δlv	1-Month London Interbank Offered Rate (%)
51	3MLIBOR	FRED	Δlv	3-Month London Interbank Offered Rate (%)
52	6MLIBOR	FRED	Δlv	6-Month London Interbank Offered Rate (%)
53	1YLIBOR	FRED	Δlv	One-Year London Interbank Offered Rate (%)
54	1MEuro-FF	FRED	lv	1-Month Eurodollar Deposits (London Bid) (% P.A.) minus Fed Funds
55	3MEuro-FF	FRED	lv	3-Month Eurodollar Deposits (London Bid) (% P.A.) minus Fed Funds

No.	Short Name	Source	Tran	Description
56	6MEuro-FF	FRED	<i>lv</i>	6-Month Eurodollar Deposits (London Bid) (% P.A.) minus Fed Funds
57	APFNF-AANF	Data Stream	<i>lv</i>	1-Month A2/P2/F2 Nonfinancial Commercial Paper (NCP) (% P.A.) minus 1-Month Aa NCP (% P.A.)
58	APFNF-AAF	Data Stream	<i>lv</i>	1-Month A2/P2/F2 NCP (% P.A.) minus 1-Month Aa Financial Commercial Paper (% P.A.)
59	TED	Data Stream, FRED	<i>lv</i>	3Month Tbill minus 3-Month London Interbank Offered Rate (%)
60	MAaa-10YTB	Data Stream	<i>lv</i>	Moody Seasoned Aaa Corporate Bond Yield (% P.A.) minus Y10-Tbond
61	MBaa-10YTB	Data Stream	<i>lv</i>	Moody Seasoned Baa Corporate Bond Yield (% P.A.) minus Y10-Tbond
62	MLA-10YTB	Data Stream, FRED	<i>lv</i>	Merrill Lynch Corporate Bonds: A Rated: Effective Yield (%) minus Y10-Tbond
63	MLAA-10YTB	Data Stream, FRED	<i>lv</i>	Merrill Lynch Corporate Bonds: Aa Rated: Effective Yield (%) minus Y10-Tbond
64	MLAAA-10YTB	Data Stream, FRED	<i>lv</i>	Merrill Lynch Corporate Bonds: Aaa Rated: Effective Yield (%) minus Y10-Tbond
Group 4: Treasuries				
65	FRFEDFD	Data Stream	Δlv	US FED FUNDS EFF RATE (D) - MIDDLE RATE
66	FRTBS3M	Data Stream	Δlv	US T-BILL SEC MARKET 3 MONTH (D) - MIDDLE RATE
67	FRTBS6M	Data Stream	Δlv	US T-BILL SEC MARKET 6 MONTH (D) - MIDDLE RATE
68	FRTCM1Y	Data Stream	Δlv	US TREASURY CONST MAT 1 YEAR (D) - MIDDLE RATE
69	FRTCM10	Data Stream	Δlv	US TREASURY CONST MAT 10 YEAR (D) - MIDDLE RATE
70	6MTB-FF	Data Stream	<i>lv</i>	6-month treasury bill market bid yield at constant maturity (%) minus Fed Funds
71	1YTB-FF	Data Stream	<i>lv</i>	1-year treasury bill yield at constant maturity (% P.A.) minus Fed Funds
72	10YTB-FF	Data Stream	<i>lv</i>	10-year treasury bond yield at constant maturity (% P.A.) minus Fed Funds
73	6MTB-3MTB	Data Stream	<i>lv</i>	6-month treasury bill yield at constant maturity (% P.A.) minus 3M-Tbills
74	1YTB-3MTB	Data Stream	<i>lv</i>	1-year treasury bill yield at constant maturity (% P.A.) minus 3M-Tbills
75	10YTB-3MTB	Data Stream	<i>lv</i>	10-year treasury bond yield at constant maturity (% P.A.) minus 3M-Tbills
76	BKEVEN05	FRB	<i>lv</i>	US Inflation compensation: continuously compounded zero-coupon yield: five-year (%)
77	BKEVEN10	FRB	<i>lv</i>	US Inflation compensation: continuously compounded zero-coupon yield: 10-year (%)
78	BKEVEN1F4	FRB	<i>lv</i>	BKEVEN1F4
79	BKEVEN1F9	FRB	<i>lv</i>	BKEVEN1F9
80	BKEVEN5F5	FRB	<i>lv</i>	US Inflation compensation: coupon equivalent forward rate: 5-10 years (%)
Group 5: Foreign Exchange (FX)				
81	US_CWBN	Data Stream	Δln	US NOMINAL DOLLAR BROAD INDEX - EXCHANGE INDEX
82	US_CWMN	Data Stream	Δln	US NOMINAL DOLLAR MAJOR CURR INDEX - EXCHANGE INDEX
83	US_CSFR2	Data Stream	Δln	CANADIAN \$ TO US \$ NOON NY - EXCHANGE RATE
84	EU_USFR2	Data Stream	Δln	EURO TO US\$ NOON NY - EXCHANGE RATE
85	US_YFR2	Data Stream	Δln	JAPANESE YEN TO US \$ NOON NY - EXCHANGE RATE
86	US_SFFR2	Data Stream	Δln	SWISS FRANC TO US \$ NOON NY - EXCHANGE RATE
87	US_UKFR2	Data Stream	Δln	UK POUND TO US \$ NOON NY - EXCHANGE RATE

From Daily to Quarterly Factors: Weighting Schemes After I obtain daily financial factors $G_{D,t}$, I use weighting schemes proposed in the literature on Mixed Data Sampling (MIDAS) regressions to form quarterly factors, denoted $G_{D,t}^Q$. Let G_t^D denote a factor in daily frequency formed from the daily financial dataset, and let G_t^Q denote a quarterly aggregate of the corresponding daily factor time series. Let $G_{ND-j,d_t,t}^D$ denote the value of a daily factor on the j -th day counting backwards from the survey deadline d_t in quarter t . Hence, the day d_t of quarter t corresponds to $j = 0$, so the daily factor on the survey deadline is $G_{ND,d_t,t}^D$. For simplicity, I suppress the subscript d_t , writing $G_{ND-j,t}^D$.

I compute the quarterly aggregate of a daily financial factor as a weighted average of observations over the ND business days before the survey deadline. This means that the forecaster's information set includes daily financial data up to the previous ND business days before the survey deadline. The quarterly factor G_t^Q is defined as:

$$G_t^Q(w) = \sum_{j=1}^{ND} w_j \times G_{ND-j,t}^D$$

where w_j is a weight. I consider the following three types of weighting schemes to convert daily factor observations to

quarterly aggregates. Each weighting scheme weights information by some function of the number of days prior to the survey deadline.

1. $w_i = 1$ for $i = 1$ and $w_i = 0$ otherwise. This weighting scheme places all weight on the data from the last business day before the survey deadline and zero weight on any data prior to that day.
2. $w_i = \delta^i / \sum_{j=1}^{ND} \delta^j$, where I consider a range of δ values with $\delta \in \{0.1, 0.2, 0.3, 0.7, 0.8, 0.9, 1.0\}$. The smaller the δ , the more rapidly information prior to the survey deadline is down-weighted. This down-weighting is progressive but not non-monotonic. The case $\delta = 1$ corresponds to a simple average of observations across all days.
3. The third parameterization uses two parameters $\theta = (\theta_1, \theta_2)'$ and allows for non-monotonic weighting of past information. The weights are defined as:

$$w(i; \theta_1, \theta_2) = \frac{f\left(\frac{i}{ND}; \theta_1, \theta_2\right)}{\sum_{j=1}^{ND} f\left(\frac{j}{ND}; \theta_1, \theta_2\right)}$$

where $f(x; a, b) = x^{a-1}(1-x)^{b-1} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}$, and $\Gamma(a)$ is the gamma function $\Gamma(a) = \int_0^\infty x^{a-1}e^{-x} dx$. The weights $w(i; \theta_1, \theta_2)$ are the Beta polynomial MIDAS weights of Ghysels et al. (2007), based on the Beta function. This weighting scheme is flexible enough to generate a wide range of possible shapes with only two parameters.

I consider these possible weighting schemes and choose the optimal weighting scheme w^* from 24 candidate weighting schemes for each daily financial factor G_t^D by minimizing the sum of squared residuals in a regression of $y_{j,t+h}$ on G_t^Q :

$$y_{j,t+h} = \alpha + \beta \times G_t^Q(w) + u_{t+h}$$

This procedure is conducted in real time using recursive regressions. I re-estimate the weights at each date in the sample recursively over time, using the entire history of data available in real time prior to each out-of-sample forecast. I assume that $ND = 14$, which implies that forecasters use daily information from at most the past two weeks before the survey deadline. This process is repeated for each daily financial factor in $G_{D,t}$ to form quarterly factors $G_{D,t}^Q$.

OD.2.7 LDA Data

The LDA data are used as inputs into the machine learning forecasts. The database for my Latent Dirichlet Allocation (LDA) analysis contains around one million articles published in *Wall Street Journal* between January 1984 to Dec 2023. The current vintage of the results reported here is based a randomly selected sub-sample of 200,000 articles over the same period, one-fifth size of the entire database. The sample selection procedure follows Bybee et al. (2021). First, I remove all articles prior to January 1984 and after June 2022 and exclude articles published in weekends. Second, I exclude articles with subject tags associated with obviously non-economic content such as sports. Third, I exclude articles with the certain headline patterns, such as those associated with data tables or those corresponding to regular sports, leisure, or books columns. I filter the articles using the same list of exclusions provided by Bybee et al. (2021). Last, I exclude articles with less than 100 words.

Processing of texts The processing of the texts can be summarized into five steps:

1. Tokenization: parse each article’s text into a white-space-separated word list retaining the article’s word ordering.
2. I drop all non-alphabetical characters and set the remaining characters to lower-case, remove words with less than 3 letters, and remove common stop words and URL-based terms. I use a standard list of stop words from the Python library *gensim.parsing.preprocessing*.
3. Lemmatization and Stemming: lemmatization returns the original form of a word using external dictionary *Textblob.Word* in Python and based on the context of the word. For instance, as a verb, “went” is converted to “go”. Stemming usually refers to a heuristic process that removes the trailing letters at the end of the words, such as from “assesses” to “assess”, and “really” to “real”. I use the Python library *Textblob.Word* to implement the lemmatization and *SnowballStemmer* for the stemming. The results are not very sensitive to the particular Python packages being used.
4. From the first three steps, I obtain a list of uni-grams which are a list of singular words. For example, “united” and “states” are uni-grams from “united states”. From the list of uni-grams, I generate a set of bi-grams as all pairs of (ordered) adjacent uni-grams. For example, “united states” together is one bi-gram. I then exclude uni-grams and bi-grams appearing in less than 0.1% of articles.
5. Last, I convert an article’s word list into a vector of counts for each uni-gram and bi-gram. For example, the vector of counts [5, 7, 2] corresponds to the number of times the words [“federal”, “reserve”, “bank”] appear in the article.

The LDA Model The LDA model Blei et al. (2003) essentially achieves substantial dimension reduction of the word distribution of each article using the following assumptions. I assume a factor structure on the vectors of word counts. Each factor is a topic and each article is a parametric distribution of topics, specified as follows,

$$\underbrace{w_i}_{\text{word dist of article } i}^{V \times 1} \sim \text{Mult} \left(\underbrace{\Phi'}_{\text{topic-word dist.}}, \underbrace{\theta_i}_{\text{topic dist.}}^{K \times 1}, \underbrace{N_i}_{\text{\# of words}} \right)$$

where Mult is the multinomial distribution. In the above equation, w_i is a vector of word counts of each unique term (uni-gram or bi-gram) in article i , whose size is equal to the number of unique terms V . K is the number of factors in article i . In the estimation, I assume $K = 180$ following Bybee et al. (2021). Φ is a matrix sized $K \times V$, whose k th row and v th column is equal to the probability of the unique term v showing up in topic k . θ_i stores the weights of all k topics contained in article i , which sum up to one. Dimension reduction is achieved as long as $K \ll V$ (the number of topics are significantly smaller than the number of unique terms). More specifically, it reduces the dimension from $T \times V$ to $T \times K$ (the size of θ) + $K \times V$ (the size of Φ).

Real-time news factors. I also generate real-time news factors for each month t starting from January 1991. In theory, I could train the LDA model using each real-time monthly vintage but it is computationally challenging. Instead, I simplify the procedure by training the LDA model using quarterly vintages $t, t+3, t+6$, etc, and use the LDA model parameters estimated at t to filter news paper articles within the quarter and generate news factors for those months. More specifically, given every article's word distribution $w_{i,t+s}$, for $s = 0, 1, 2$, and the estimated real-time topic-word distribution parameters $\hat{\Phi}_t$ using articles till date t , one can obtain the filtered topic distribution of each article $\hat{\theta}_{i,t+s}$, as follows,

$$\underbrace{w_{i,t+s}}_{\text{word dist of article } i \text{ at time } t+s}^{V \times 1} \sim \text{Mult} \left(\underbrace{\hat{\Phi}'}_{\text{topic-word dist.}}, \underbrace{\hat{\theta}_{i,t+s}}_{\text{topic dist.}}^{K \times 1}, \underbrace{N_{i,t+s}}_{\text{\# of words}} \right).$$

LDA Estimation I use the built-in LDA model estimation toolbox in the Python library <https://pypi.org/project/gensim/> to implement the model estimation. The model requires the following initial inputs and parameters and it is estimated using Bayesian methods. In theory, maximum-likelihood estimation is possible but it is computationally challenging.

1. I create a document-term matrix $\mathbf{W}$ as a collection of w_i for all articles i in the sample. The number of rows in $\mathbf{W}$ is equal to the number of articles in my sample and the number of columns in $\mathbf{W}$ is equal to the number of unique uni-gram and bi-grams (after being filtered) across all articles. The matrix $\mathbf{W}$ is used as an input for the LDA model estimation. I then follow Bybee et al. (2021) and set the number of topics K to be 180. The authors used Bayesian criteria to find 180 to be an optimal number of topics.
2. In the Python library Gensim, the key parameters of the LDA estimation are α and β . With a higher value of α , the documents are composed of more topics. With a higher value of β , each topic contains more terms (uni- or bi-grams). In the implementations, I do not impose any explicit restrictions on initial values of those parameters and set them to be "auto". These two parameters, alongside Φ' and $\{\theta_i\}_i$, are estimated by the toolbox from Python library <https://pypi.org/project/gensim/>.

Real-time LDA Factors With the estimated topic weights $\theta_{i,t}$ of each article i from the LDA model, I further construct time series of the overall news attention to each topic, or a news factor. The value of the topic k at time t is the average weights of topic k of all articles published at t , specified as follows,

$$F_{k,t} = \frac{\sum_i \hat{\theta}_{i,k,t}}{\text{\# of articles at } t}$$

for all topics k . I construct daily LDA factors by aggregating all articles published on each calendar day. The value of topic k at day t is the average weights of topic k across all articles published that day.

OD.2.8 Machine Variables to Be forecast

Returns and price growth When evaluating the MSE ratio of the machine relative to that of a benchmark survey, I use the machine forecast for the return or price growth measure that most closely corresponds to the concept that survey respondents are asked to predict:

1. CFO survey asks respondents about their expectations for the S&P 500 return over the next 12 months. Following Nagel and Xu (2021), I interpret the survey to be asking about $r_{t,t+12}^d$, the one-year CRSP value-weighted return (including dividends) from the current survey month to the same month one year ahead.
2. Gallup/UBS survey respondents report the return (including dividends) they expect on their own portfolio one year ahead. I interpret the survey to be asking about $r_{t,t+12}^d$, the one-year CRSP value-weighted return (including dividends) from the current survey month to the same month one year ahead.
3. Livingston survey respondents provide 12-month ahead forecasts of the S&P 500 index. I convert the level forecast to price growth forecast by taking the log difference between the 12-month ahead level forecast and the nowcast of the S&P 500 index for the current survey month. Therefore, I interpret the survey to be asking about the one-year price growth in the S&P 500 index.
4. Bloomberg Consensus Forecasts asks survey respondents about the end-of-year closing value of the S&P 500 index. I interpret the survey to be asking about the h -month price growth in the S&P 500 index. The horizon of the forecast changes depending on when in the year the panelists are answering the survey.
5. Michigan Survey of Consumers (SOC) asks respondents about their perceived probability that an investment in a diversified stock fund would increase in value in the year ahead. I interpret the question to be asking about the one-year price growth in the S&P 500 index.
6. Conference Board (CB) survey asks respondents about their categorical belief on whether they expect stock prices to increase, decrease, or stay the same over the next year. I interpret the question to be asking about the one-year price growth in the S&P 500 index.

Earnings growth (IBES “Street” Earnings) For earnings growth forecasts, I use a quarterly S&P 500 total earnings series based on IBES street earnings per share (EPS), as described above. Street earnings exclude discontinued operations, extraordinary charges, and other non-operating items, making them better aligned with the earnings measure targeted by survey respondents. I convert EPS to total earnings using the S&P 500 index divisor and use the resulting quarterly series directly, prior to any monthly interpolation, since the machine learning algorithm operates at a quarterly frequency. The IBES street earnings series spans 1983Q4 to 2021Q4.

For Long-Term Growth (LTG) forecasts, IBES defines LTG as the “expected annual increase in operating earnings over the company’s next full business cycle. These forecasts refer to a period of between three to five years.” I compare survey responses of LTG against machine forecasts under alternative interpretations of LTG. First, I consider machine forecasts of annual five-year forward growth, i.e., annual earnings growth from four to five years ahead (Bianchi et al. (2024)). Second, I consider machine forecasts of annualized five-year growth, i.e., annual earnings growth from current quarter to five years ahead, following the interpretation in Bordalo et al. (2019). Third, I consider machine forecasts of annualized earnings growth from one to 10 years ahead, following the interpretation in Nagel and Xu (2021).

Inflation I construct forecasts of annual inflation defined as $\pi_{t+4,t} = \log\left(\frac{PGDP_{t+4}}{PGDP_t}\right)$, where $PGDP_t$ is the quarterly level of the chain-weighted GDP price index. Following Coibion and Gorodnichenko (2015), I use the vintage of inflation data that is available four quarters after the period being forecast.

OD.2.9 Machine Input Data: Predictor Variables

The vector $\mathbf{Z}_{jt} \equiv (y_{j,t}, \hat{\mathbf{G}}'_t, \mathbf{W}'_{jt})'$ is an $r = 1 + r_G + r_W$ vector which collects the data at time t with

$$\mathbf{Z}_{jt} \equiv (y_{j,t}, \dots, y_{j,t-p_y}, \hat{\mathbf{G}}'_t, \dots, \hat{\mathbf{G}}'_{t-p_G}, \mathbf{W}'_{jt}, \dots, \mathbf{W}'_{jt-p_W})'$$

a vector of contemporaneous and lagged values of $\mathbf{Z}_{jt}$, where p_y, p_G, p_W denote the total number of lags of $y_{j,t}, \hat{\mathbf{G}}'_t, \mathbf{W}'_{jt}$, respectively. The predictors below are listed as elements of $y_{j,t}, \hat{\mathbf{G}}'_{jt}$, or $\mathbf{W}'_{jt}$ for variables.

Stock return and price growth predictor variables and specifications For y_j equal to CRSP value-weighted returns or S&P 500 price index growth, I first predict the one-year log stock return or price growth that is expected to occur h quarters into the future from time $t+h-4$ to $t+h$, i.e., $\mathbb{E}_t[r_{t+h-4,t+h}]$. For horizons longer than one year, since the h -quarter long horizon return is the sum of one-year returns between time t to $t+h$, I first forecast the forward one-year returns separately and then add the components together to get machine forecasts of h -quarter long horizon returns. The forecasting model considers the following variables. Lags of the dependent variable:

1. y_{t-1}, y_{t-2} one and two quarter lagged stock returns or price growth.

The factors in $\hat{\mathbf{G}}'_{jt}$ are formed from three large datasets separately:

1. $\mathbf{G}_{M,t-k}$, for $k = 0, 1$ are factors formed from a real-time macro dataset $\mathcal{D}^M$ with 92 real-time macro series; includes both monthly and quarterly series, with monthly series converted to quarterly according to the method described in the data appendix.
2. $\mathbf{G}_{F,t-k}$, for $k = 0, 1$ are factors formed from a financial data set $\mathcal{D}^F$ with 147 monthly financial series.
3. $\mathbf{G}_{D,t-k}^Q$, for $k = 0$ are quarterly factors formed from a daily financial dataset $\mathcal{D}^D$ of 87 daily financial indicators. The raw daily series are first converted to daily factors $\mathbf{G}_{D,t}(\mathbf{w})$ and the daily factors are aggregated up to quarterly observations $\mathbf{G}_{D,t}^Q(\mathbf{w})$ using a weighted average of daily factors, with the weights $\mathbf{w}$ dependent on two free parameters that are chosen to minimize the sum of squared residuals in a regression of $y_{j,t+h}$ on $\mathbf{G}_{D,t}(\mathbf{w})$.

The variables in $\mathbf{W}'_{jt}$ include:

1. *LDA topics* $F_{k,t-j}$, for topic $k = 1, 2, \dots, 50$ and $j = 0, 1$. The value of the topic k at time t is the average weights of topic k of all articles published at t .
2. *Macro data surprises* from the money market survey. The macro news include, GDP growth (Q/Q percentage change), core CPI (Month/Month change), unemployment rate (percentage point), and nonfarm payroll (month/month change). I include first release, second release, and final release for GDP growth. This constitutes six macro data surprises per quarter.
3. *FOMC surprises* are defined as the changes in the current-month, 1, 2, 6, 12, and 24 month-ahead federal funds futures (FFF) contract rate and the changes in the 1, 2, 4, and 8 quarter-ahead Eurodollar (ED) futures contracts, from 10 minutes before to 20 minutes after each FOMC announcement. When benchmarking against a survey, I use the last FOMC meeting before the survey deadline to compute surprises. For surveys that do not have a clear deadline, I compute surprises using the last FOMC in the first month of the quarter. When benchmarking against moving average, I use the last FOMC meeting before the end of the first month in each quarter to compute surprises. This leaves 10 FOMC surprise variables per quarter.
4. *Stock market jumps* are accumulated 30-minute window negative and positive jumps in the S&P 500 around news events over the previous quarter.
5. *Long-term growth of earnings*: five-year growth of the SP500 earnings per share.
6. *Short rates*. When forecasting returns or price growth, the machine controls for the current nominal short rate, $\log(1 + 3MTB_t/100)$, imposing a unit coefficient. This is equivalent to forecasting the future return minus the current short rate.

The 92 macro series in $\mathcal{D}^M$ are selected to represent broad categories of macroeconomic time series. The majority of these are real activity measures: real output and income, employment and hours, consumer spending, housing starts, orders and unfilled orders, compensation and labor costs, and capacity utilization measures. The dataset also includes commodity and price indexes and a handful of bond and stock market indexes, and foreign exchange measures. The financial dataset $\mathcal{D}^F$ is an updated monthly version of the of 147 variables comprised solely of financial market time series used in Ludvigson and Ng (2007). These data include valuation ratios such as the dividend-price ratio and earnings-price ratio, growth rates of aggregate dividends and prices, default and term spreads, yields on corporate bonds of different ratings grades, yields on Treasuries and yield spreads, and a broad cross-section of industry, size, book-market, and momentum portfolio equity returns. A detailed description of the series is given in the Data Appendix of the online supplementary file at www.sydneyludvigson.com/s/ucc_data_appendix.pdf. The 87 daily financial indicators in $\mathcal{D}^D$ include daily time series on commodities spot prices and futures prices, aggregate stock market indexes, volatility indexes, credit spreads and yield spreads, and exchange rates.

Earning growth predictor variables and specifications For y_t equal to S&P 500 log earning growth, I construct a forecasted value for y_t , denoted $\hat{y}_{t|t-h}$, based on information known up to time t using the following variables. Lags of the dependent variable:

1. y_{t-1}, y_{t-2} one and two quarter lagged earnings growth.

The factors in $\hat{\mathbf{G}}'_{jt}$ are formed from three large datasets separately:

1. $\mathbf{G}_{M,t-k}$, for $k = 0, 1$ are factors formed from a real-time macro dataset $\mathcal{D}^M$ with 92 real-time macro series; includes both monthly and quarterly series, with monthly series converted to quarterly according to the method described in the data appendix.
2. $\mathbf{G}_{F,t-k}$, for $k = 0, 1$ are factors formed from a financial data set $\mathcal{D}^F$ with 147 monthly financial series.
3. $\mathbf{G}_{D,t-k}^Q$, for $k = 0$ are quarterly factors formed from a daily financial dataset $\mathcal{D}^D$ of 87 daily financial indicators. The raw daily series are first converted to daily factors $\mathbf{G}_{D,t}(\mathbf{w})$ and the daily factors are aggregated up to quarterly observations $\mathbf{G}_{D,t}^Q(\mathbf{w})$ using a weighted average of daily factors, with the weights $\mathbf{w}$ dependent on two free parameters that are chosen to minimize the sum of squared residuals in a regression of $y_{j,t}$ on $\mathbf{G}_{D,t}(\mathbf{w})$.

The variables in $\mathbf{W}'_{jt}$ include:

1. *LDA factors* $F_{k,t-j}$, for topic $k = 1, 2, \dots, 50$ and $j = 0, 1$. The value of the topic k at time t is the average weights of topic k of all articles published at t .
2. *Macro data surprises* from the money market survey. The macro news include, GDP growth (Q/Q percentage change), core CPI (Month/Month change), unemployment rate (percentage point), and nonfarm payroll (month/month change). I include first release, second release, and final release for GDP growth. This constitutes six macro data surprises per quarter.
3. *FOMC surprises* are defined as the changes in the current-month, 1, 2, 6, 12, and 24 month-ahead federal funds futures (FFF) contract rate and the changes in the 1, 2, 4, and 8 quarter-ahead Eurodollar (ED) futures contracts, from 10 minutes before to 20 minutes after each FOMC announcement. When benchmarking against a survey, I use the last FOMC meeting before the survey deadline to compute surprises. For surveys that do not have a clear deadline, I compute surprises using the last FOMC in the first month of the quarter. When benchmarking against moving average, I use the last FOMC meeting before the end of the first month in each quarter to compute surprises. This leaves 10 FOMC surprise variables per quarter.
4. *Stock market jumps* are accumulated 30-minute window negative and positive jumps in the S&P 500 around news events over the previous quarter.

Inflation predictor variables For y_j equal to inflation, the forecasting model considers the following variables. Lags of the dependent variable:

1. $y_{t-1,t-h-1}$ one quarter lagged inflation.

The factors in $\hat{\mathbf{G}}'_{jt}$ are formed from three large datasets separately:

1. $\mathbf{G}_{M,t-k}$, for $k = 0, 1$ are factors formed from a real-time macro dataset $\mathcal{D}^M$ with 92 real-time macro series; includes both monthly and quarterly series, with monthly series converted to quarterly according to the method described in the data appendix.
2. $\mathbf{G}_{F,t-k}$, for $k = 0, 1$ are factors formed from a financial data set $\mathcal{D}^F$ with 147 monthly financial series.
3. $\mathbf{G}_{D,t-k}^Q$, for $k = 0$ are quarterly factors formed from a daily financial dataset $\mathcal{D}^D$ of 87 daily financial indicators. The raw daily series are first converted to daily factors $\mathbf{G}_{D,t}(\mathbf{w})$ and the daily factors are aggregated up to quarterly observations $\mathbf{G}_{D,t}^Q(\mathbf{w})$ using a weighted average of daily factors, with the weights $\mathbf{w}$ dependent on two free parameters that are chosen to minimize the sum of squared residuals in a regression of $y_{j,t+h}$ on $\mathbf{G}_{D,t}(\mathbf{w})$.

The variables in $\mathbf{W}'_{jt}$ include:

1. $\mathbb{F}_{jt-k}^{(i)}[y_{jt+h-k}]$, lagged values of the i th type's forecast, where $k = 1, 2$
2. $\mathbb{F}_{jt-1}^{(s \neq i)}[y_{jt+h-1}]$, lagged values of other type's forecasts, $s \neq i$
3. $var_N \left(\mathbb{F}_{t-1}^{(\cdot)}[y_{jt+h-1}] \right)$, where $var_N(\cdot)$ denotes the cross-sectional variance of lagged survey forecasts
4. $skew_N \left(\mathbb{F}_{t-1}^{(\cdot)}[y_{jt+h-1}] \right)$, where $skew_N(\cdot)$ denotes the cross-sectional skewness of lagged survey forecasts

5. Trend inflation measured as $\bar{\pi}_{t-1} = \begin{cases} \rho\bar{\pi}_{t-2} + (1-\rho)\pi_{t-1}, \rho = 0.95 & \text{if } t < 1991\text{Q4} \\ \text{CPI10}_{t-1} & \text{if } t \geq 1991\text{Q4} \end{cases}$, where CPI10 is the median SPF forecast of annualized average inflation over the current and next nine years. Trend inflation is intended to capture long-run trends. When long-run forecasts of inflation are not available, as is the case pre-1991Q4, I use a moving average of past inflation.
6. $\dot{GDP}_{t-1}$ = detrended gross domestic product, defined as the residual from a regression of GDP_{t-1} on a constant and the four most recent values of GDP as of date $t-8$. See Hamilton (2018).
7. $\dot{EMP}_{t-1}$ = detrended employment, defined as the residual from a regression of EMP_{t-1} on a constant and the four most recent values of EMP as of date $t-8$. See Hamilton (2018).
8. $\mathbb{N}_t^{(i)}[\pi_{t,t-h}]$ = Nowcast as of time t of the i th percentile of inflation over the period $t-h$ to t .

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